

Grating Coupler Design Based on Silicon-On-Insulator

by

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B.Sc., Shenzhen University, 2011

A THESIS SUBMITTED IN PARTIAL FULFILLMENT OF
THE REQUIREMENTS FOR THE DEGREE OF

MASTER OF APPLIED SCIENCE

in

The Faculty of Graduate Studies

(Electrical and Computer Engineering)

THE UNIVERSITY OF BRITISH COLUMBIA

(Vancouver)

April 2013

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Abstract

Silicon-on-insulator has become a promising platform for high-density integrated photonics circuits. The large refractive index contrast between the functional silicon layer and its cladding raises a coupling issue between an optical fibre and on-chip devices. Grating coupler provides a compact and efficient way to tackle the coupling issue between the optical fibre and silicon waveguide. In this thesis, a universal design methodology, which accommodates various etch depths, silicon thicknesses, and cladding materials has been demonstrated and verified by both FDTD simulation and measurement results. A fully etched grating coupler with a sub-wavelength grating structure has been proposed to reduce the large back reflection of existing fully etched grating couplers. Back reflection of the proposed fully etched grating coupler has been reduced from more than 20% to about 5%. The insertion loss and bandwidth of the proposed structure have also been improved. In addition, a bidirectional grating coupler for vertical coupling has been proposed to improve the insertion loss and bandwidth of the traditional grating coupler. A simulated insertion loss of -1.5dB with a 3dB bandwidth of more 100nm has been achieved with the proposed structure.

Preface

I am the first author on two conference papers and titled “Fully Etched Grating Coupler With Low Back Reflection” and “ Universal Grating Coupler Design” respectively. I also co-authored five papers, including two journal papers [44, 45] and three conference papers. In addition, I am the main author of a book chapter [9]. During the last year, I proposed a universal design methodology to design grating couplers for various fabrication processes and applications. This design flow has been implemented in *Pyxis* to automatically generate the desired grating coupler with user-specified input parameters. The validity of this design method has been verified by both theoretical calculation and measurement results. I also proposed a fully etched grating coupler with sub-wavelength grating structure to reduce the back reflection of the existing fully etched grating coupler. In addition, I proposed a bidirectional grating coupler structure for vertical coupling, the potential of which has been verified by theoretical calculation and numerical simulation.

My complete list of publications are:

1. Yun Wang, Jonas Flueckiger, Charlie Lin, and Lukas Chrostowski, “ Fully Etched Grating Coupler With Low Back Reflection ” Photonics North 2013 (accepted);
2. Yun Wang, Jonas Flueckiger, Charlie Lin, and Lukas Chrostowski, “ Universal Grating Coupler Design ” Photonics North 2013 (accepted);
3. Wei Shi, Han Yun, Wen Zhang, Charlie Lin, Ting Kai Chang, Yun Wang, Nicolas A. F. Jaeger, and Lukas Chrostowski . “ Ultra-Compact, High-Q Silicon Micodisk Reflectors ”, Optics Express, Vol.20, Issue 20, pp.21846(2012)

4. Wei Shi, Han Yun, Charlie Lin, Mark Greenburg, Xu Wang, Yun Wang, Sahba Talebi Fard, Jonas Flueckiger, Nicolas A. F. Jaeger, and Lukas Chrostowski, “ Ultra-compact, flat-top demultiplexer using anti-reflection contra-directional couplers for CWDM networks on silicon ” Optics Express, (accepted),
<http://www.opticsinfobase.org/oe/upcomingissue.cfm>
5. Wei Shi, Ting Kai Chang, Han Yun, Wen Zhang, Yun Wang, Charlie Lin, Nicolas A. F. Jaeger, and Lukas Chrostowski, “ Differential Measurement of Transmission Losses of Integrated Optical Components Using Waveguide Ring Resonators ” Proc. SPIE 8412, Photonics North 2012, 84120R (October 23, 2012); doi:10.1117/12.2001409
6. Wei Shi, Han Yun, Charlie Lin, Xu Wang, Yun Wang, Jonas Flueckiger, Nicolas A. F. Jaeger, and Lukas Chrostowski, “ Silicon CWDM Demultiplexers Using Contra-Directional Couplers ”, CLEO, 2013
7. Han Yun, Wei Shi, Yun Wang, Lukas Chrostowski, and Nicolas A.F. Jaeger “2x2 Adiabatic 3-dB Couplign on Silicon-on-insulator Rib Waveguides ”, Photonics North 2013 (accepted);
8. Lukas Chrostowski and Michael Hochberg, “ Silicon Photonics Design ” (Chapter 6), 2013, ISBN: 9781105948749

Table of Contents

Abstract	ii
Preface	iii
Table of Contents	v
List of Tables	vi
List of Figures	vii
Acknowledgements	viii
Dedication	ix
1 Introduction	1
1.1 Silicon Photonics	1
1.2 Challenge of Coupling Light Into Nanophotonic Waveguide	3
1.3 Grating Coupler	5
1.4 Slab Waveguide and Channel Waveguide	6
1.5 Polarization of Waveguide Modes	7
1.6 State-of-the-art Grating Couplers	8
1.7 Measurement Setup	12
2 Theory And Numerical Methods	14
2.1 Bragg Condition	14
2.1.1 Bragg's Law	14
2.1.2 The Bragg Condition for Grating Coupler	15
2.2 Effective Index Method	17

Table of Contents

2.3	Finite-Difference Time-Domain Method	21
3	Detuned Grating Coupler	25
3.1	Detuned Shallow Etched Grating Coupler	25
3.1.1	Initial Condition	26
3.1.2	Design Parameters	27
3.1.3	Optimization of the Grating Coupler	34
3.1.4	Design Stability	38
3.2	Universal Grating Coupler Design	41
3.2.1	Design Approach	41
3.2.2	Simulation Results	43
3.2.3	Mask Layout	49
3.2.4	Measurement Results	50
4	Fully Etched Grating Coupler	56
4.1	Regular Fully Etched Grating Coupler	56
4.1.1	Bottlenecks of Regular Fully Etched Grating Couplers	57
4.1.2	Optimization of the Regular Fully Etched Grating Cou- plers	60
4.2	Improved Fully Etched Grating Coupler	64
4.2.1	Design Approach	65
4.2.2	Simulation Results	67
4.2.3	Measurement Results	68
5	Vertical Grating Coupler	70
5.1	Regular Vertical Grating Couplers	70
5.2	Bidirectional Grating Coupler	73
5.2.1	Device Layout	73
5.2.2	Design and Simulation	74
6	Discussion And Future Work	78
	Bibliography	81

Table of Contents

Appendices

A FDTD code to generate universal grating coupler model 89

B Pyxis code for universal grating coupler design 101

List of Tables

1.1	State-of-the-art grating couplers	11
3.1	Initial values	26
3.2	Tuning coefficients of various parameters	32
3.3	Input paramters	44
3.4	Tuning coefficient of different parameters	51
6.1	Comparison of the published results and the result of the grating coupler generated by the universal design methodology	79

List of Figures

1.1	Schematic of SOI wafer	2
1.2	Schematic of grating coupler	4
1.3	Schematic of slab waveguide and channel waveguide in SOI	6
1.4	(a) The amplitude of the electric field of the first order TE-like mode in a rectangular channel waveguide; (b) The amplitude of the magnetic field of the first order TE-like mode in a rectangular channel waveguide [9]	9
1.5	(a)The amplitude of the electric field of the first order TM-like mode in a rectangular channel waveguide; (b)The amplitude of the magnetic filed of the first order TM-like mode in a rectangular channel waveguide [9]	10
1.6	(a) Illustration of the automated setup; (b) automated setup	12
1.7	Fibre array ribbon, ribbon holder and ribbon arm [26]	13
2.1	Bragg's Law	15
2.2	Schematic of a grating coupler	16
2.3	Diagram for Bragg condition	17
2.4	Cross section of silicon-on-insulator waveguide	18
2.5	Schematic of a SOI strip waveguide	18
2.6	Schematic of the effective index of a strip waveguide	19
2.7	Effective index of TE and TM modes	21
2.8	FDTD mesh for a grating coupler	22
2.9	(a) Schematic of simulation structure for input grating coupler; (b) Schematic of simulation structure for output grating coupler	24

List of Figures

3.1	Insertion loss and back reflection to the waveguide of the initial grating coupler	27
3.2	Variations of grating period	28
3.3	Variations of duty cycle	29
3.4	Variations of etch depth	30
3.5	Variations of incident angle	31
3.6	Schematic of reflections at different interfaces	32
3.7	Variations for BOX thickness	33
3.8	Variations for cladding thickness	34
3.9	Grating coupler design fabricated through OpSIS-IME	35
3.10	Images of fibre ribbon	36
3.11	Impacts of the gap between fibre ribbon tip and photonic chip on the insertion loss and bandwidth of the grating coupler . .	37
3.12	Spectra of simulation results with different gap distance and measurement results	38
3.13	The insertion losses of the same grating coupler design at different positions of the chip	39
3.14	The 3dB bandwidths of the same grating coupler design at different positions of the chip	40
3.15	The central wavelengths of the same grating coupler design at different positions of the chip	40
3.16	Flow chart of the universal design method	42
3.17	Universal grating couplers with 10 degree incident angle for TE mode wave with oxide cladding	46
3.18	Universal grating couplers with 10 degree incident angle for TM mode wave with oxide cladding	46
3.19	Universal grating coupler with 10 degree incident angle for TE mode wave with air cladding	47
3.20	Universal grating coupler with 10 degree incident angle for TM mode wave with air cladding	47
3.21	Comparison of designs generated by universal grating coupler model and the optimized design for 1550nm TE wave with 10 degree incident angle	48

List of Figures

3.22	Mask layout of a grating coupler with focusing grating curve	50
3.23	Measurement vs. simulation of universal grating couplers with 10 degree incident angle	52
3.24	Measurement vs. simulation of universal grating couplers with 15 degree incident angle	52
3.25	Measurement vs. simulation of universal grating couplers with 20 degree incident angle	53
3.26	(a) Peak power of simulation and measurement results with 10 incident angle; (b) comparison of simulated and measured wavelength mismatch with 10 degree incident angle	54
3.27	(a) Peak power of simulation and measurement results with 15 incident angle; (b) comparison of simulated and measured wavelength mismatch with 15 degree incident angle	54
3.28	(a) Peak power of simulation and measurement results with 20 incident angle; (b) comparison of simulated and measured wavelength mismatch with 20 degree incident angle	55
4.1	Schematic of fully etched grating coupler	57
4.2	Schematic of diffraction, reflection and penetration of a reg- ular fully etched grating coupler	58
4.3	Insertion loss of grating couplers with 20 degree incident angle for TE operation wave as function of the thickness of the buried oxide.	59
4.4	Directionality, insertion loss and reflection to waveguide of a general fully-etched grating coupler	61
4.5	Mask layout of a fully etched grating coupler test structure .	62
4.6	Comparison of measurement result and simulation result of fully-etched grating coupler	63
4.7	Comparison of back reflections between shallow-etched grat- ing coupler and fully-etched grating coupler	64
4.8	Schematic of a fully etched grating coupler with sub-wavelength gratings	66

List of Figures

4.9	Comparison of regular fully etched grating coupler and the fully etched grating coupler with minor sub-wavelength gratings.	67
4.10	Measurement results of the regular fully etched grating coupler and the fully-etched grating coupler with minor sub-wavelength gratings	68
5.1	Diagram of wave vectors for vertical grating coupler	71
5.2	Insertion loss and back reflection of a regular vertical grating coupler	72
5.3	Schematic of bidirectional grating coupler for vertical coupling	73
5.4	Cross section of a bidirectional grating coupler	74
5.5	The insertion loss of the optimized bidirectional grating coupler	75
5.6	Insertion of the bidirectional vertical grating coupler as function of wavelength for different offset	76
5.7	Schematic of a bidirectional grating coupler with incident wave off from the centre of the grating.	77

Acknowledgements

I would like to thank my supervisor Dr. Lukas Chrostowski. For the care he provided to help me settle down when I first came to the University of British Columbia as an international student, for the patient guidance and help he provided when I was faced with difficulties and for the friendly and warm atmosphere he has created within our research group.

I would like also to thank Dr. Nicolas Jaeger for the helpful instruction and kindly help with my research.

Also, I would like to thank my colleagues Wei Shi, Xu Wang, Han Yun, Jonas Flueckiger, Samantha Grist, Charlie Lin, and Sahba Talebi Fard. Special thanks to Wei Shi for his instructive discussions and kind help.

In addition, I would also like to acknowledge the nano-fabrication center at University of Washington, CMC Microsystems and OpSIS IME for the fabrication of my devices and Lumerical Solutions Inc. for their technical support.

Dedication

To my father Junping Wang and my mother Miaoling Zhang, who have shown me what love is. Specifically, the courage, determination and passion from my father to help me find what I truly love and to follow my own heart, and the rationality from my mother, which helped me to think independently when I met obstacles in my life. To my fiancé Ge Shi, for her dedication and love.

Chapter 1

Introduction

1.1 Silicon Photonics

After dominating the electronics industry for decades, silicon is on the verge of becoming the material of choice for the photonics industry [21]. The early work in the area of silicon photonics dates back to the late 1980s and the early 1990s [12, 20, 43, 46, 51, 52, 52, 53, 62]. The motivation for silicon photonics is its compatibility with the silicon Integrated Circuit (IC) manufacturing process, which represents the most spectacular convergence of technological sophistication and economies of scale. The industry is able to produce microprocessors with hundreds of millions of components, all integrated onto a thumb-size chip, and to offer them at extremely low price.

Another motivation is the availability of high-quality silicon-on-insulator (SOI) wafers, an ideal platform for creating planar waveguide circuits. A schematic of an SOI wafer is shown in Fig. 1.1. An SOI wafer consists of three layers: a silicon substrate at the bottom (of Fig. 1.1) for mechanical support, a buried oxide (BOX) layer (in the middle) acting as the insulator layer, and another silicon layer (on the top) acting as the functional wave guiding layer. There may be another cladding layer on top of the silicon layer for protection. The strong optical confinement offered by the high index contrast between Si ($n=3.45$) and SiO_2 ($n = 1.45$) makes it possible to make photonics devices on the scale of a few hundred nanometers. Such small dimensions are required for compatibility with the IC processing. The *Smart Cut* [2] technology commercialized by *Soitec* also propelled the commercial applications of SOI into exponential growth and entered the mainstream of Ultra Large Scale Integration (ULSI).

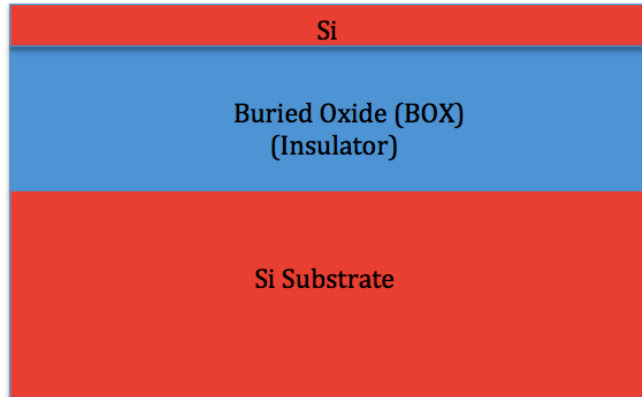


Figure 1.1: Schematic of SOI wafer

The main application envisioned for silicon photonics is optical interconnects for CMOS electronics [34]. Conventional wisdom holds that optical interconnects are much better suited than copper interconnects in handling high data rates. The physical benefits of changing the technology used for interconnects to silicon photonic chips in computing and switching systems have been detailed in [33]. With the advantages of design simplification, architectural advantages and physical benefits such as reduction of power dissipation in interconnects and voltage isolation etc., light beams that have dominated long-distance communication are progressively taking over in shorter distance networks. Luxtera announced the world's first 40 Gigabit Optical Active Cable (OAC) in 2007 [28]. Intel demonstrated their 4x12.5 Gbps CWDM silicon photonics link using integrated hybrid silicon lasers [25, 37] in 2010. IBM unveiled the holey optochip to transfer information at terabit per second speed in 2012 [10]. All those advances show the potential of silicon photonics as the future solution for optical interconnects.

Another important application of silicon photonics is bio-sensing. A disposable mass-produced sensor would be attractive as it could grow the market for biosensors. Sensor applications are somewhat different from optical communication ones as there are other low cost optical technologies that

compete in this space [23]. One likely application area for silicon photonics is the so-called lab-on-a-chip, in which both reaction and analysis are performed in a single device. In the future, this could be extended to include electronic intelligence and wireless communications to create intelligent sensor networks for environmental monitoring [21].

In addition, silicon has material properties that are important for a new class of mid-infrared (IR) photonics devices. These include the linear and non-linear optical properties in the mid-wave IR spectrum. The high optical intensity arising from the large index contrast (between *Si* and *SiO₂*) makes it possible to observe nonlinear optical interactions, such as Raman and Kerr effects, in chip-scale devices. Far from being limited to the near-IR data communication band, from 1.26 to 1.67 microns, silicon also has a low loss wavelength window extending from 1.1 to nearly 7 microns [22]. But oxide limits its use in SOI to 3 microns. Silicon has high thermal conductivity (10x higher than GaAs), high optical damage threshold (10x higher than GaAs), and high third-order optical nonlinearities [21].

1.2 Challenge of Coupling Light Into Nanophotonic Waveguide

Due to the large refractive index contrast between the silicon core ($n \approx 3.47$ at 1550 nanometers) and the silicon dioxide cladding ($n \approx 1.5$ at 1550 nanometers), propagation modes are highly confined within the waveguide with a dimension on the order of a few hundred nanometers, which enables large-scale integration. However, the small feature size of the waveguide raises the problem of huge mode mismatch between the optical mode within an optical fibre and the mode within the waveguide. The cross-sectional area of an optical fibre-core (with a diameter of 10 microns) is almost 3,000 times larger than that of a silicon waveguide (with dimensions of 500 nanometers x 220 nanometers).

Several approaches have been demonstrated to tackle the problem of mode mismatch. Edge coupling using spot size converters and lensed fi-

1.2. Challenge of Coupling Light Into Nanophotonic Waveguide

bles is one solution used to address this, and high-efficiency coupling with an insertion loss below 0.5 dB has been demonstrated [30]. However, this approach can be only used at the edge of the chips, and the implementation of such designs requires complicated post-processes, which increase the packaging cost dramatically. The alignment of such devices during the measurement requires high accuracy, which often takes a lot of effort.

The grating couplers are an alternative solution to tackle the issue of mode mismatch. Compared to the edge coupling, grating couplers have several advantages: alignment to grating couplers during measurement is much easier than alignment to edge couplers; the fabrication of grating couplers does not require post-processing, which reduces the fabrication cost; grating couplers can be put anywhere on a chip, which provides flexibility in the design as well as enabling wafer scale automated testing. Both academic and industrial research groups have demonstrated high efficiency grating couplers [32, 35, 41].

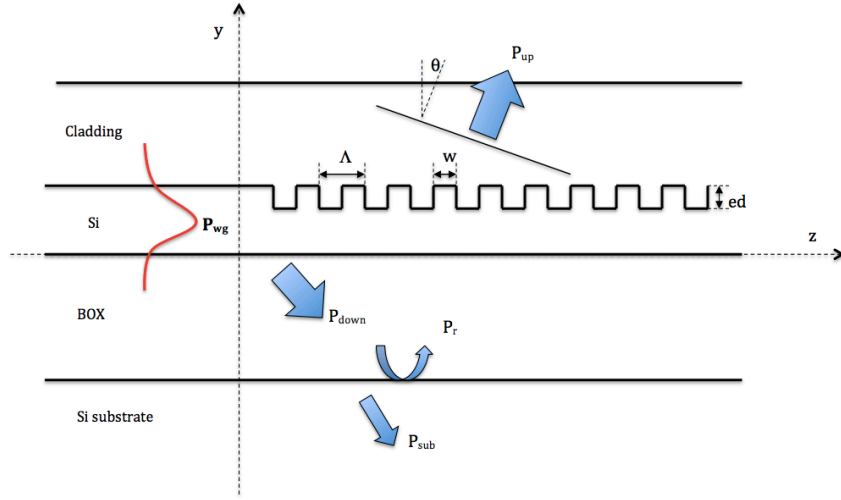


Figure 1.2: Schematic of grating coupler

1.3 Grating Coupler

Figure 1.2 is a schematic of a shallow-etched grating coupler design in silicon-on-insulator. The thickness of the functional *Si* layer and the thickness of the buried oxide (BOX) layer are determined by the wafer type. A cladding layer is often employed to protect the functional silicon layer, as shown in Fig. 1.2. However, for some uses such as bio-sensing, air cladding is required. In Fig. 1.2 :

- Λ denotes the grating period, which represents the length of the periodic pattern;
- W denotes duty cycle of the grating coupler, which is the width of the grating tooth;
- ff denotes fill factor, which is defined as the ratio of the grating period and the duty cycle, i.e, $ff = \Lambda/W$;
- θ denotes the incident angle of the grating coupler, which is the angle between the incident wave and the normal to the grating surface;
- ed denotes the heights of the grating teeth, which are defined as etch depth.

In addition to the geometric variables defined above, some other terms are often used when discussing grating couplers:

1. Directionality: the ratio between the power diffracted upwards (P_{up}) and the input power from the waveguide (P_{wg}) [35], which is usually expressed in decibels (dB) as $10 \cdot \log_{10}(P_{up}/P_{wg})$;
2. Insertion loss (coupling efficiency): the ratio between the power coupled into the fundamental mode of the fibre (P_{fund}) and the input power from the waveguide (P_{wg}), which is usually expressed in dB as $10 \cdot \log_{10}(P_{fund}/P_{wg})$;
3. Penetration loss: the ratio between the power lost in the substrate (P_{sub}) and the input power from the waveguide, which is usually expressed in dB as $10 \cdot \log_{10}(P_{sub}/P_{wg})$;

4. Back reflection to the waveguide: due to the refractive index contrast between the silicon wire waveguide and the grating, part of the input light from the waveguide will be reflected back into the waveguide. The ratio between the reflected power and the input power from the waveguide is called back reflection to the waveguide. It is usually expressed in dB or in percentage. This back reflection is unwanted because it will cause Fabry-Perot oscillations by reflecting back and forth between the input and output grating couplers [35, 55].

1.4 Slab Waveguide and Channel Waveguide

At angles of incidence above the critical angle it is possible to achieve total internal reflection from a dielectric interface given that the incident electromagnetic wave is in the medium with the higher refractive index. So, it should be possible to place a slab of dielectric with a high refractive index between two media with lower refractive indices and confine a plane wave to propagate in the high index dielectric slab [19]. Such a waveguide is called the slab waveguide. A schematic of a slab waveguide in SOI is shown in the left side of Fig. 1.3. The slab is such that it extends to infinity in the x and z directions. However, the waveguide used in photonic systems are the ones with two-dimensional refractive index profiles, which is shown in the right side of Fig. 1.3. Such a waveguide is called the channel waveguide.

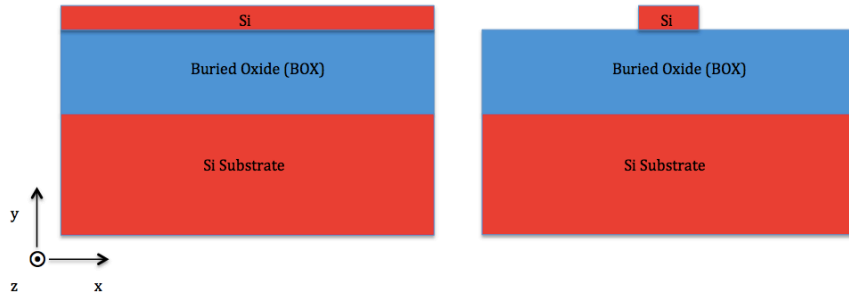


Figure 1.3: Schematic of slab waveguide and channel waveguide in SOI

1.5 Polarization of Waveguide Modes

If we orient our coordinate system so that the interface between the silicon and the cladding and the interface between the silicon and the buried oxide lie in the xz plane (shown in Fig. 1.3) and the guided mode propagates along the z axis, then the wave vectors for the incident, reflected, and transmitted waves are all contained in the yz plane. Thus the yz plane is called the plane of incidence [19]. Transverse electric (TE) modes are defined as these modes having their electric fields perpendicular to the plane of incidence (i.e. x axis here) and transverse magnetic (TM) modes are defined as these modes having their magnetic fields perpendicular to the plane of incidence. The amplitude of the electric field and magnetic field of the fundamental (first order or lowest order) TE-like mode are shown in Fig. 1.4. The amplitude of the electric field and magnetic field of the fundamental TM-like mode are shown in Fig. 1.5. By comparing the amplitude of the fundamental TE-like mode and the fundamental TM-like mode, we can see that the TE-like mode is better confined than the TM-like mode. In all subsequent sections of this thesis, channel waveguides are assumed, and we will drop the -like extension and refer to the modes as simple TE and TM modes.

The propagation constant, β , is employed to represent the behaviour of different modes within the channel waveguide and is defined as [19]:

$$\beta = \frac{2 \cdot \pi}{\lambda_0} \cdot n_{\text{eff}} \quad (1.1)$$

where λ_0 denotes the operation wavelength and n_{eff} denotes the effective refractive index of the mode. The effective refractive index is introduced to describe and compare the confined modes within the channel waveguide and is defined as:

$$n_{\text{eff}} = \frac{c}{v_p} = \frac{\beta}{k_0} \quad (1.2)$$

where c is the speed of light in vacuum, v_p is the phase velocity of the mode, and k_0 denotes the wave vector in free space, i.e., $k_0 = 2\pi/\lambda$.

1.6 State-of-the-art Grating Couplers

Some state-of-the-art grating couplers are listed in Table 1.1. Important parameters such as central wavelength, polarization, insertion loss, and bandwidth, for comparison purpose, are shown in the table. Corresponding fabrication details are also listed because the performance of a grating coupler is highly dependent on how it is made. The insertion loss of the grating coupler is mainly determined by the thickness of the top silicon layer and the thickness of the buried oxide (BOX), because these two values affect the phase conditions of different wavelengths at the interface between the grating layer and the buried oxide layer. Bottom distributed Bragg reflectors (DBR) [31] and top overlays [56] have been used to decrease the insertion loss, but non-standard wafers and fabrication processes are required for these approaches. Both shallow etched and fully etched grating couplers are shown in the table and the central wavelengths are around 1550 nm and 1310 nm, which are in the two most commonly used optical windows in telecommunications. These grating couplers listed are mainly designed for TE mode operation.

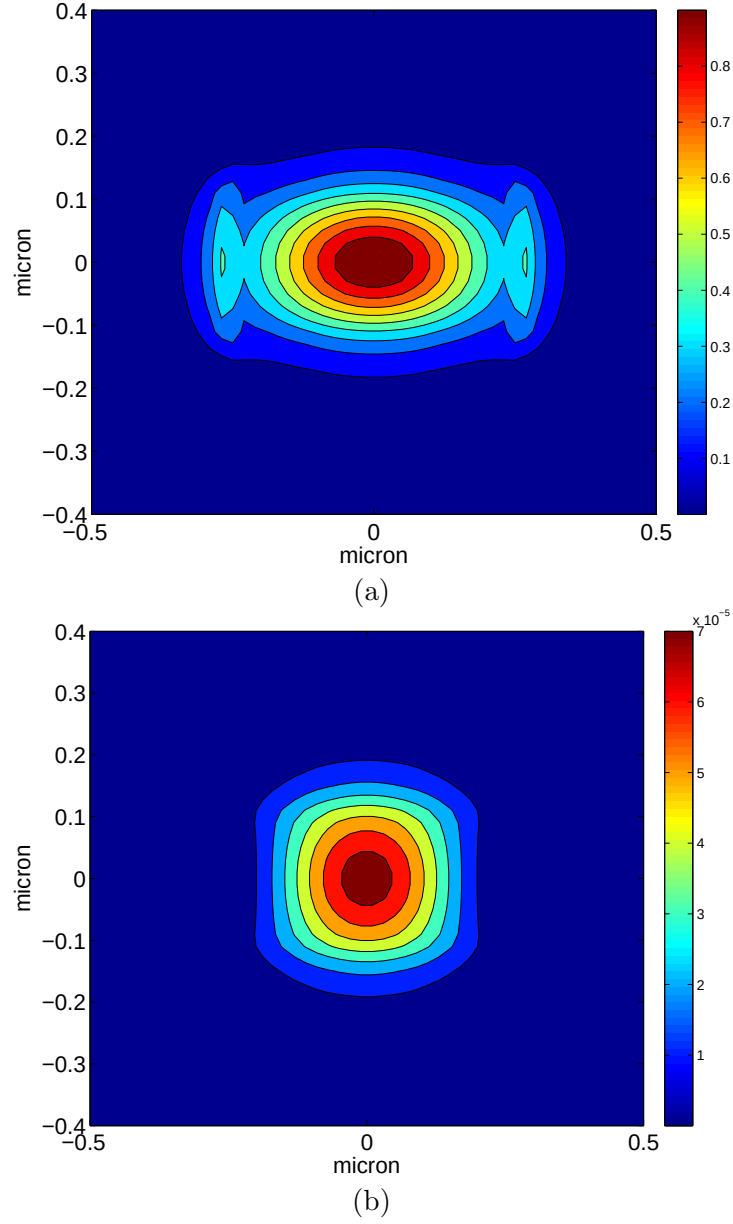


Figure 1.4: (a) The amplitude of the electric field of the first order TE-like mode in a rectangular channel waveguide; (b) The amplitude of the magnetic field of the first order TE-like mode in a rectangular channel waveguide [9]

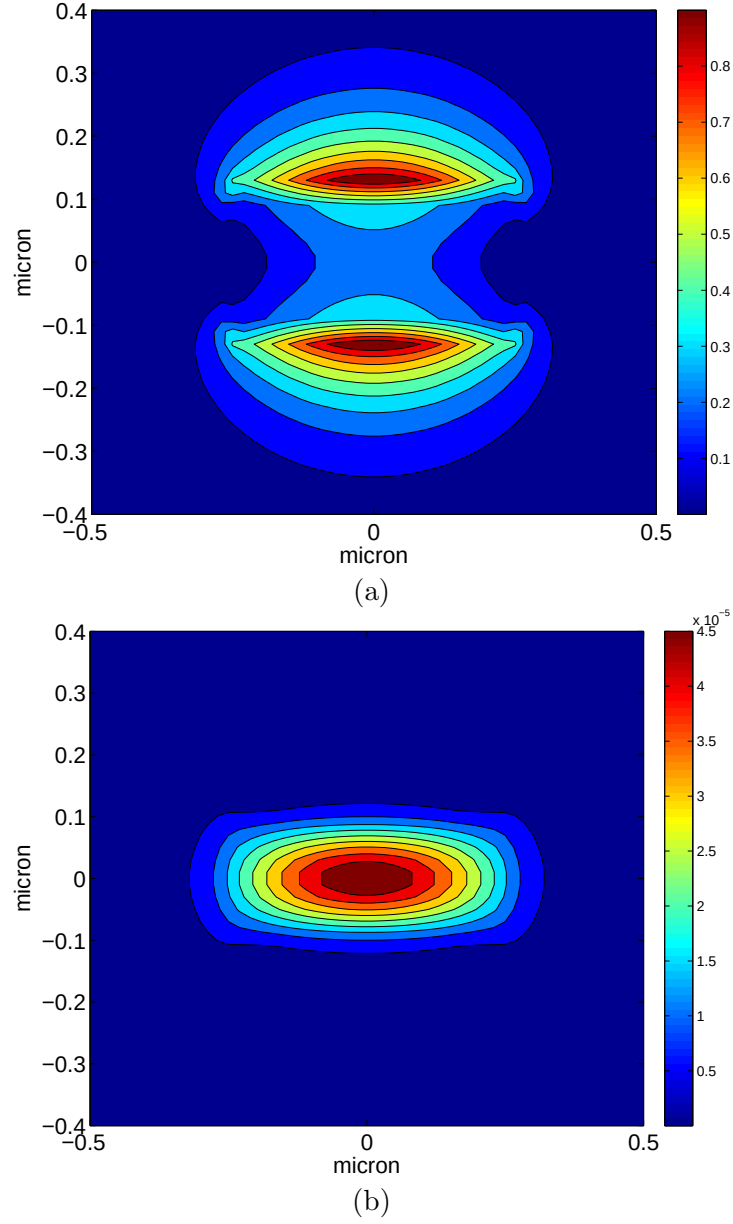
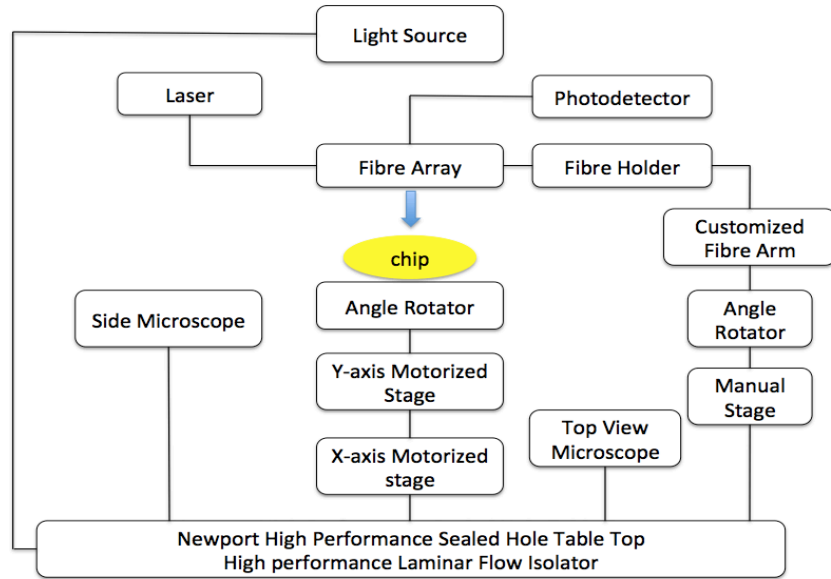


Figure 1.5: (a)The amplitude of the electric field of the first order TM-like mode in a rectangular channel waveguide; (b)The amplitude of the magnetic field of the first order TM-like mode in a rectangular channel waveguide [9]

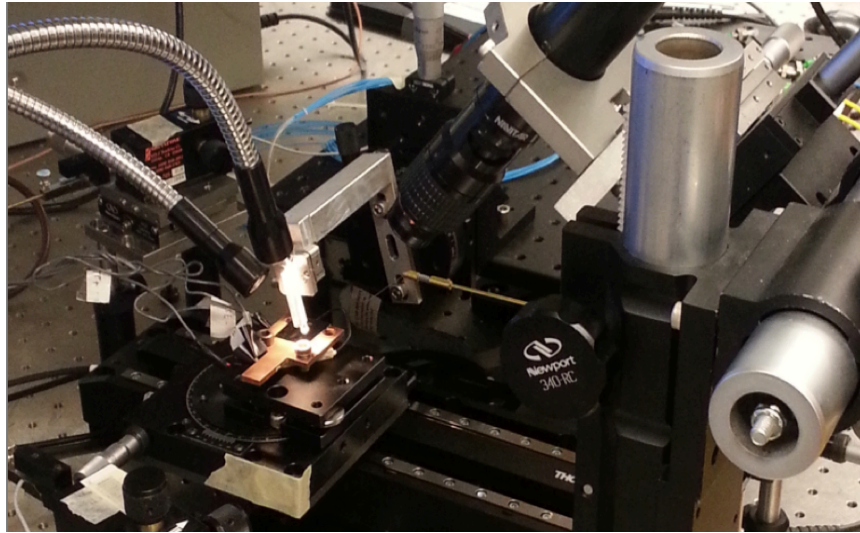
	Wavelength	Polarization	Insertion Loss	Bandwidth	Process
2006 [48]	1550nm	TE	-5.1 dB	40nm (1dB)	220nm Si, 1um BOX, shallow etch
2010 [56]	1530nm	TE	-1.6dB	80nm (3dB)	amorphous Si overlay, shallow etch
2010 [3]	1530nm	TE	-1.2dB	N/A	340nm Si, 2um BOX, shallow etch
2011 [35]	1310nm	TE	-3dB	58nm	400nm Si, shallow etch
2012 [60]	1550nm	TE	-4.4dB	45nm (1.5dB)	220nm Si, 2um BOX, shallow etch
2012 [31]	1490nm	TE	-0.75 dB	N/A	shallow etch, bottom DBR
2010 [13]	1550nm	TM	-3.7dB	60nm	full etch
2012 [58]	1550nm	TE	-2.29dB	60nm	full etch, 250 Si ,1um BOX

Table 1.1: State-of-the-art grating couplers

1.7 Measurement Setup



(a)



(b)

Figure 1.6: (a) Illustration of the automated setup; (b) automated setup

1.7. Measurement Setup

The automated setup used for the measurement of the devices shown in this thesis are built by Charlie Lin [26] and Jonas Flueckiger. The illustration and physical map of the automated setup are shown in Fig. 1.6 (a) and Fig. 1.6(b), respectively. The chip is placed on a platform which is on top of an angle rotator. An X-axis motorized stage and a Y-axis stage are seated below the angle rotator, which forms a cross configuration. The fibre array ribbon is held by a custom-made aluminum fibre ribbon holder, which is suspended on top of the chip platform. An aluminum arm is used to hold the fibre ribbon holder and is attached to an angle rotator (shown in Fig. 1.7). The angle rotator is fixed onto a Z-axis actuator that is bolted to a raised platform so the fibre ribbon height can be manually adjusted accordingly. The ribbon-to-chip image is captured two microscopes; one microscope shows the top view and is used for alignment purpose; the other microscope is angled from the side to display the height displacement between the fibre array and the chip to prevent crashing the fibre array into the chip during alignment. The light source for the microscopes illuminates the chip platform at an angle from behind the fibre ribbon [26].

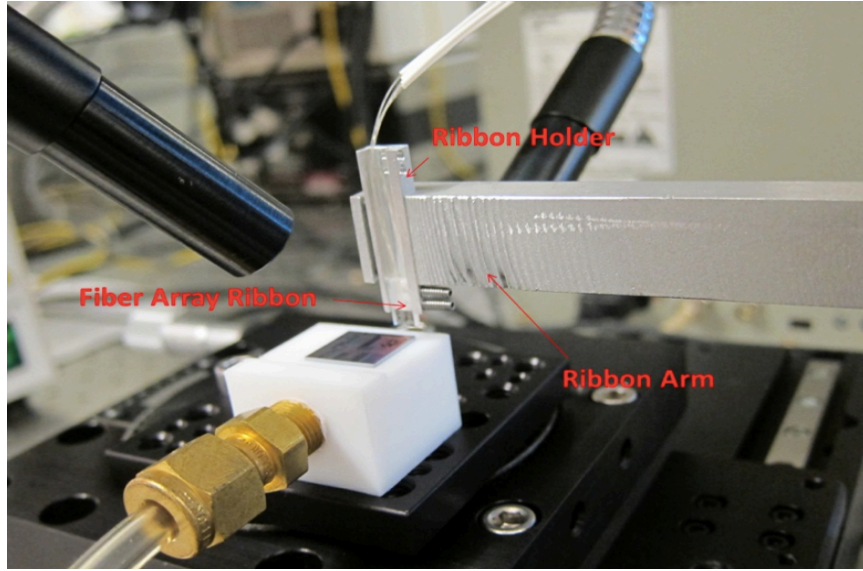


Figure 1.7: Fibre array ribbon, ribbon holder and ribbon arm [26]

Chapter 2

Theory And Numerical Methods

In this chapter, we will give a short overview on the fundamental theories and numerical methods related to grating coupler design. We will start from Bragg's Law, from which we will derive the most fundamental theory to deal with periodic structures, i.e., the Bragg condition. Also, the effective Index Method (EIM) will be introduced to obtain an approximation of the effective index of refraction of a slab waveguide. Finally, the Finite-Difference Time Domain (FDTD) method is introduced to obtain the optimized solution for the grating coupler design and to predict the performance.

2.1 Bragg Condition

2.1.1 Bragg's Law

A schematic of Bragg's Law is shown in Fig. 2.1. Periodic dots are seated in the air with a period of d in the y direction. A plane wave is incident on the periodic structure and is scattered by each plane periodic dots in such a way that the portion scattered from the second plane of dots undergoes an extra length of $2d\sin(\theta)$, as compared to the portion scattered by the first plane of dots. Depending on the phase condition, either constructive interference or destructive interference occurs. Constructive interference occurs when the extra length is equal to an integer multiple of the wavelength of the incident wave, i.e.:

$$2 \cdot d \cdot \sin\theta = m \cdot \lambda \quad (2.1)$$

2.1. Bragg Condition

where m is an integer, λ is the wavelength of the incident wave, and θ is the scattering angle.

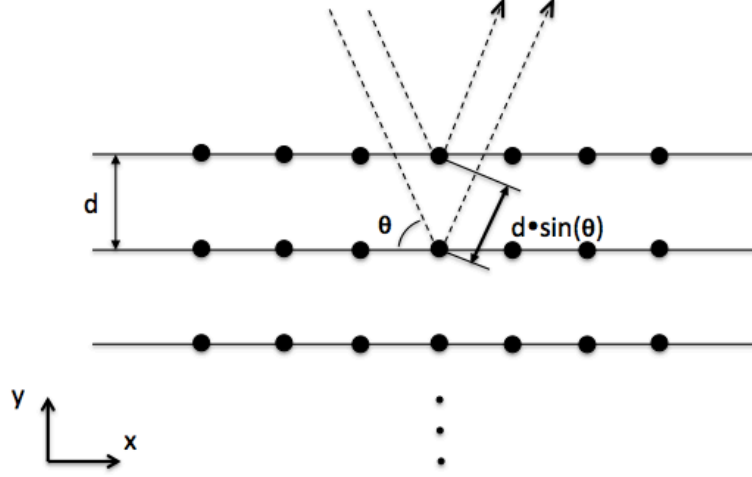


Figure 2.1: Bragg's Law

2.1.2 The Bragg Condition for Grating Coupler

The most fundamental formula concerning periodic structures is Bragg condition. The grating coupler discussed in this thesis is a one-dimensional periodic structure, as shown in Fig. 2.2. In Fig. 2.2 we assume that the wave incident on the grating is a guided wave propagating in a slab waveguide, and its direction of propagation is in the same plane as the grating and is normal to the grating teeth. The general form of the Bragg condition can be expressed as:

$$\beta - k_z = m \cdot K \quad (2.2)$$

where β denotes the wave vector of the input wave, i.e., $\beta = 2\pi n_{wg}/\lambda$, n_{wg} denotes the effective index of the incident wave, k_z denotes the component of the wave vector of the diffracted wave in the direction of the incident wave, where $k_z = 2\pi n_c/\lambda$; and $K = 2\pi/\Lambda$, which is determined by the periodicity

2.1. Bragg Condition

of the structure. This relation can also be depicted by a diagram shown in Fig. 2.3, which is easier to understand.

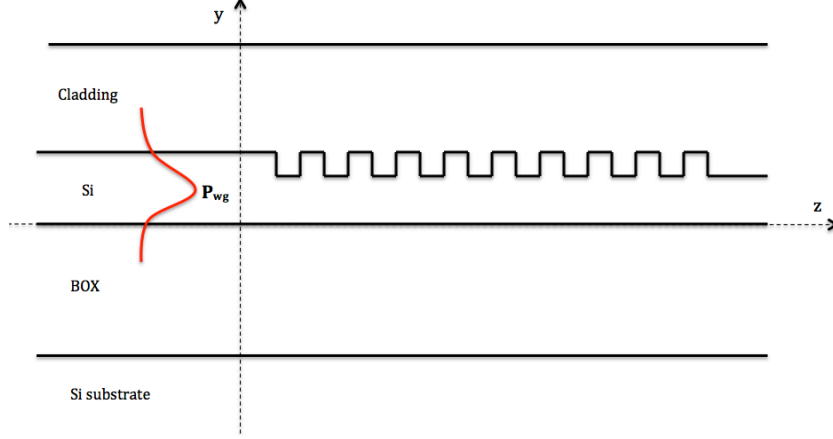


Figure 2.2: Schematic of a grating coupler

The diffractions of a grating coupler can be observed in those directions where constructive interference is achieved. Each value of m that results in diffraction is referred to as the m -th order diffraction:

$$n_{\text{eff}} \cdot \Lambda - n_c \cdot \Lambda \cdot \sin\theta = m \cdot \lambda \quad (2.3)$$

where n_{eff} denotes the effective index of the grating, n_c denotes the effective index of the fibre mode in the cladding, Λ denotes the period of the grating, θ denotes the diffraction angle, λ denotes the wavelength of the incident wave (or out-coupled wave), and m is an integer denoting the diffraction order. The diffraction order normally used for coupling is the first order (i.e., $m = 1$), so the Bragg Condition for a grating coupler can be further simplified to:

$$n_{\text{eff}} - n_c \cdot \sin(\theta) = \frac{\lambda}{\Lambda} \quad (2.4)$$

It should be noted that the Bragg condition is only exact for infinite gratings, i.e., one-dimensional grating with infinite grating period. In a

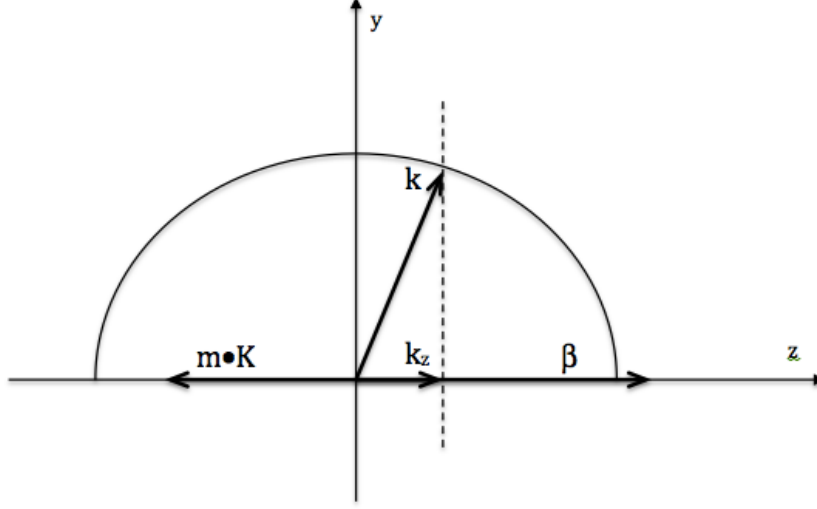


Figure 2.3: Diagram for Bragg condition

finite grating, there is not a discrete k -vector for which diffraction occurs, but a range of k -vectors around the one predicted by the Bragg condition [49].

2.2 Effective Index Method

The Effective Index Method (EIM) was initially proposed for the analysis of dielectric waveguides with rectangular cores [24]. The basic idea of this method is to replace a two-dimensional waveguide with a one-dimensional one with an effective index derived from the geometry and refractive index of the original structure [7]. The EIM has been applied to various structures such as optical waveguides [17, 29], optical fibres [7], and waveguide arrays [8].

Figure 2.4 shows the schematic of a silicon-on-insulator waveguide. It is a three-layer structure with SiO_2 ($n=1.444$) in the middle and Si ($n=3.47$) layers on the top and bottom. Air ($n=1$) has been employed as the cladding in this structure. The procedure for calculating the effective index for silicon-

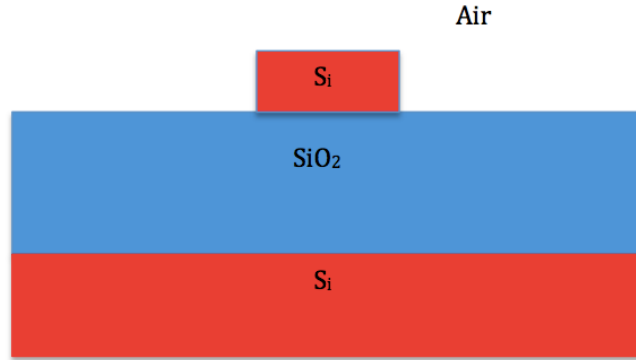


Figure 2.4: Cross section of silicon-on-insulator waveguide

on-insulator waveguide with a two-dimensional cross-section is shown in Fig. 2.5 and Fig. 2.6. The basic approach is to solve the mode condition for a particular mode type in one dimension and find the propagation constant. The effective index can be derived from the propagation constant, $n_{\text{eff}} = \beta/k_0$, and then applied to the other dimension of the structure [19].

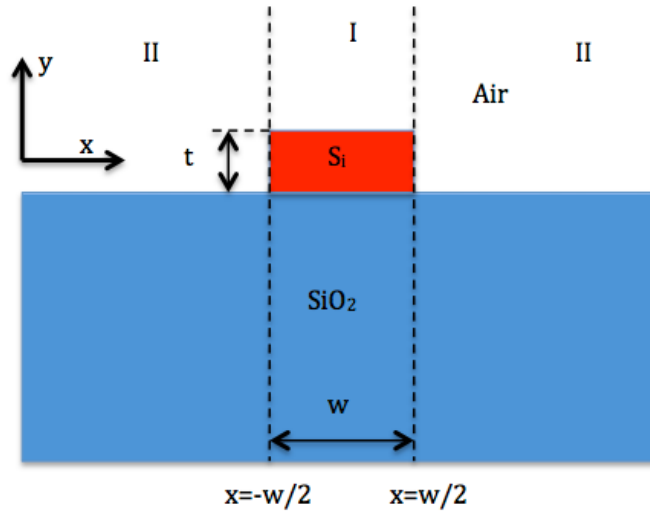


Figure 2.5: Schematic of a SOI strip waveguide

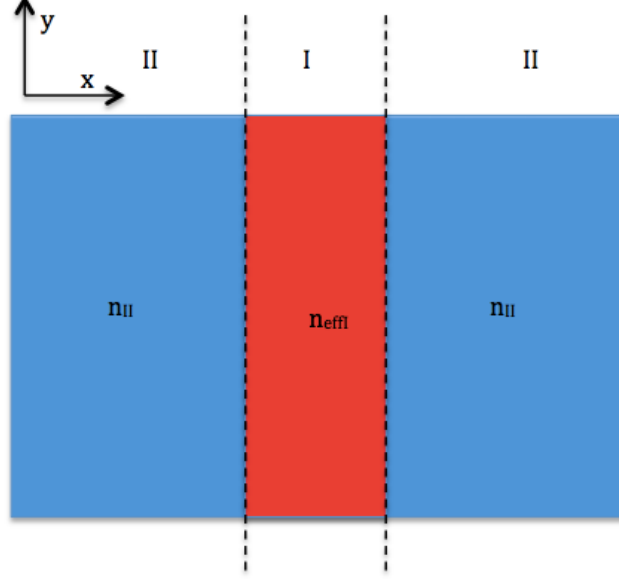


Figure 2.6: Schematic of the effective index of a strip waveguide

To begin with, we divide the structure into three regions. These three regions are $x < -w/2$, $-w/2 < x < w/2$, and $w/2 < x$, where w denote the width of the silicon waveguide. The cladding of the waveguide is air, which has a refractive index of 1. Next we decide on the mode type that we wish to solve for. For the lowest order TE-like mode, the TE_0 mode, we solve the mode condition for TE modes in the y -direction of a slab waveguide with the refractive index profile in region I . The mode condition for TE modes within region I can be expressed by [19]:

$$ht = m\pi + \tan^{-1}\left(\frac{q}{h}\right) + \tan^{-1}\left(\frac{p}{h}\right); \quad m = 0, 1, 2, \dots \quad (2.5)$$

$$h = \sqrt{k_0^2 n_2^2 - \beta^2} \quad (2.6)$$

$$p = \sqrt{\beta^2 - k_0^2 n_3^2} \quad (2.7)$$

$$q = \sqrt{\beta^2 - k_0^2 n_1^2} \quad (2.8)$$

2.2. Effective Index Method

where m is the mode order, n_1 , n_2 , n_3 , are the refractive index of the cladding, silicon core and SiO_2 , respectively. β is the propagation constant of the mode supported by the slab waveguide, which is defined as $k_0 n_{\text{eff}}$, where n_{eff} is the effective index of the mode within the slab waveguide in region I .

Now we create the one-dimensional waveguide structure shown in Fig. 2.6, in which our calculated n_{eff} is used for the central slab, i.e., for region I . The refractive index chosen for region II depends on the actual structure; nevertheless, for SOI strip waveguides with air cladding, one can use refractive index of the air, i.e., $n=1$. In the x-direction, the mode condition is solved for the appropriate TM mode of the structure. The condition for TM modes can be expressed as [19]:

$$ht = m\pi + \tan^{-1}\left(\frac{\bar{q}}{h}\right) + \tan^{-1}\left(\frac{\bar{p}}{h}\right); \quad m = 0, 1, 2, \dots \quad (2.9)$$

$$\bar{p} = \frac{n_2^2}{n_3^2} \cdot p \quad (2.10)$$

$$\bar{q} = \frac{n_2^2}{n_1^2} \cdot q \quad (2.11)$$

Using the EIM, we have now successfully converted a two-dimensional waveguide into a one-dimensional structure, and then solved for that structure. By doing this, the EIM enables us to simulate three-dimensional structures as two-dimensional ones, which saves significant computational effort and time. Following on this idea, we can take complex three-dimensional waveguides and reduce them to two-dimensional systems in which each waveguide is replaced by its effective index in a plane that is parallel to the the interface of the Si layer and the BOX layer.

The effective index of refraction of a strip waveguide, with a dimension of 220 nm by 500 nm, for both TE_0 and TM_0 modes, as a function of wavelength, are shown in Fig. 2.7. The TE_0 mode has a larger effective index than the TM_0 mode because the TE mode is better confined within the waveguide. In this thesis, we only use the first step of the EIM to find the effective index of the slab waveguide which forms the grating coupler.

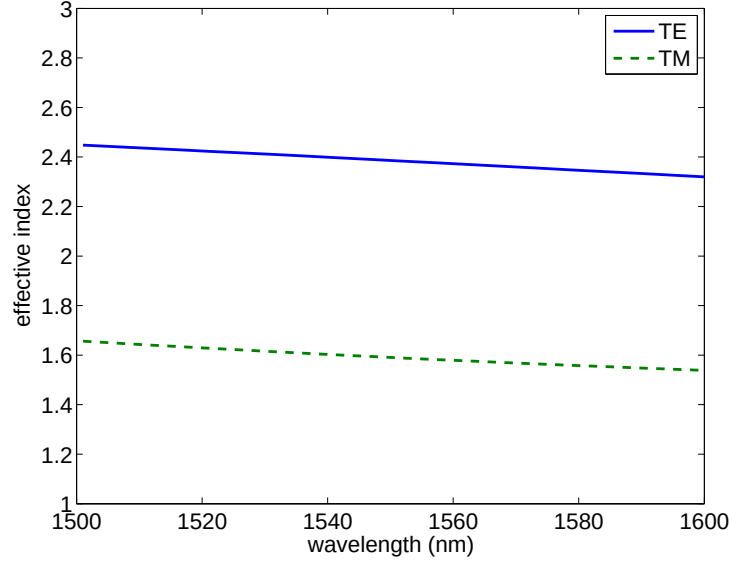


Figure 2.7: Effective index of TE and TM modes

2.3 Finite-Difference Time-Domain Method

The components used in photonic integrated circuits are normally complicated three-dimensional structures such as gratings, rings, waveguide couplers, etc. It is not possible to obtain the exact analytical solutions for these structures, except for some special cases. In practice, we use numerical methods to obtain the solutions for such structures. Finite-Difference Time-Domain (FDTD) method is a very popular numerical method used for obtaining solutions to two-dimensional and three-dimensional structures.

When Maxwell's differential equations are examined, it can be seen that the change in the E-field in time (the time derivative) depends on the change in the H-field in space (the curl). This results in the basic FDTD time-stepping relation that, at any point in space, the updated value of the E-field in time depends on the stored value of the E-field and the numerical curl of the local distribution of the H-field in space[59]. The H-field is time-stepped in a similar manner. At any point in space, the updated value of the H-field in time is dependent on the stored value of the H-field and the numerical

2.3. Finite-Difference Time-Domain Method

curl of the local distribution of the E-field in space. Iterating the E-field and H-field updates results in a “marching-in-time” process, wherein sampled-data analogs of the continuous electromagnetic waves under consideration propagate in a numerical grid stored in the computer memory[18].

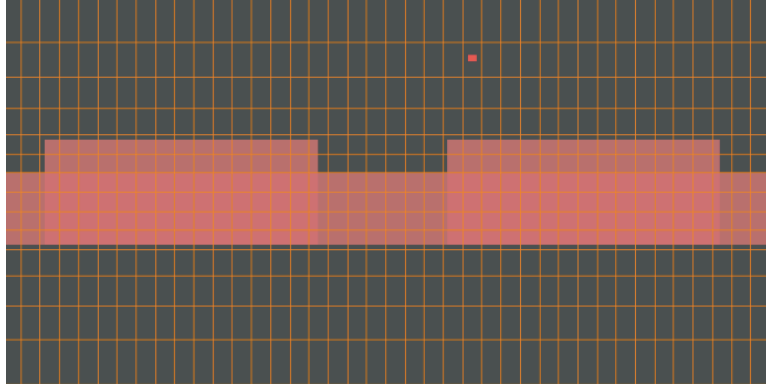


Figure 2.8: FDTD mesh for a grating coupler

During the simulation, the structure is discretized using a uniform grid as shown in Fig. 2.8. A brute-force calculation of Maxwell’s equations on each mesh point will be operated within the time domain. The simulation accuracy is highly dependent on the grid size. A smaller grid size is required to get more accurate results. The advantage of the FDTD method is that it can deal with arbitrarily complicated structures. However, the drawback of this method is long calculation times and large computational memory is required for accurate simulations. The FDTD method may not be appropriate for simulating extra long structures, but it is ideal for simulating compact structures such as grating couplers. FDTD Solutions, a commercial product from Lumerical Solutions Inc., was employed as the simulation tool for all grating couplers in this thesis.

Two-dimensional simulations are generally used to simulate grating couplers because it takes much less computational memory and simulation time. After designing a grating using 2D simulations, 3D simulations are used to verify the behaviours of the final design. The schematic shown in Fig. 2.9 depicts a general grating coupler structure: a *Si* layer on the bottom for mechanical strength, a functional *Si* layer on top of 2- μm buried oxide, and a top oxide layer for protection. The orange rectangle defines the simulation region and Perfectly Matched Layer (PML) boundary is used so that radiation appears to propagate out of the computational area and therefore, does not interfere with the fields inside. The yellow lines shown in the graphs represent frequency-domain power monitors which collect high-accuracy power flow information in the frequency domain from simulation results across spatial regions within the simulation. The green area denotes the fibre, the purpose of which will be explained below.

Two types of simulation structures are often employed to simulate a grating coupler. Figure. 2.9 (a) is used to simulate the input grating coupler and Fig. 2.9 (b) is used to simulate the output grating coupler. A polished optical fibre is presented on top of the cladding, with a light green area indicating the fibre core and a dark green area indicating the cladding of the fibre. For an input grating coupler, a fundamental TE mode is launched from the fibre core and coupled into the waveguide by the grating. Power monitors are used to record the insertion loss and reflection to the fibre of the grating coupler. For an output grating coupler, a fundamental TE mode is launched from the waveguide and out-coupled into the fibre. A mode expansion monitor is used to calculate the power that goes into the fundamental mode of the fibre [18]. This is essential for a coupler because not all of the power is coupled into the fundamental mode due to the mode mismatch.

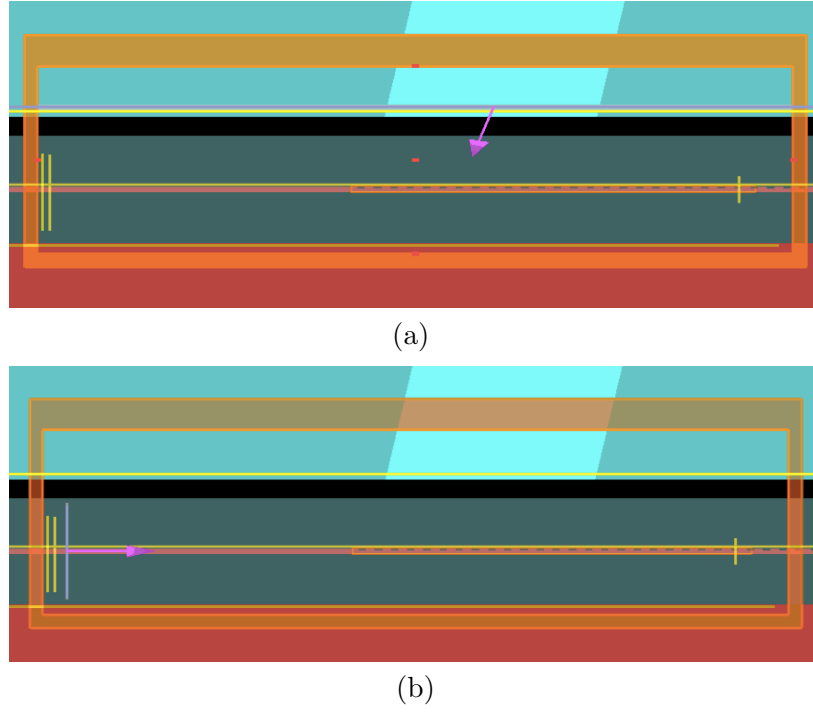


Figure 2.9: (a) Schematic of simulation structure for input grating coupler;
(b) Schematic of simulation structure for output grating coupler

Chapter 3

Detuned Grating Coupler

The most important properties for a grating coupler are the insertion loss, the back reflection to the waveguide, and the bandwidth. Compared to the fully etched grating couplers, shallow etched grating couplers have the advantages of high coupling efficiency and low back reflection to the waveguide. So it is more popularly employed in the integrated photonics circuits. A small angle is often employed between the incident wave and the normal of the grating surface, so that the large second order Bragg reflection can be avoided, which is called the detuned case.

In this chapter, the design flow for detuned shallow etched grating couplers will be presented, and the influences of various factors on the properties of the grating couplers will be discussed in detail. In addition, a universal grating coupler design methodology will be introduced to generate shallow etched grating couplers, using analytic calculations instead of numeric simulations.

3.1 Detuned Shallow Etched Grating Coupler

Designing a grating coupler follows a procedure. The first step is to get the initial condition of the desired grating coupler using theoretical calculations, and the second step is to optimize the performance of the initial design by sweeping various parameters such as grating period, duty cycle, and incident angle.

3.1.1 Initial Condition

Designing a grating coupler should follow some restrictions: some of the parameters are determined by the wafer type we use, such as the thickness of the silicon layer and the thickness of the buried oxide; some of the parameters are determined by the fabrication process, such as the cladding material, the etch depth, and the minimum feature size; and some of the parameters are decided by the application and tunability of the measurement setup, such as the central wavelength and the incident angle. In our case, all the known initial values are listed in Table 3.1.

Si	SiO_2	Cladding	Etch Depth	λ	θ
220 nm	2um	air	70 nm	1550 nm	20 degree

Table 3.1: Initial values

Given the initial values listed in Table 3.1, we can calculate the effective index of refraction of the grating. We used the Finite-difference time-domain (FDTD) method to calculate the effective index of refraction of the grating. The effective index of refraction of the shallow etched slab waveguide, n_{eff1} , is calculated to be 2.534, and the effective index of refraction of the un-etched slab waveguide, n_{eff2} , is calculated to be 2.848. Thus, the overall effective index of refraction of the grating n_{eff} can be calculated from the following equation:

$$n_{\text{eff}} = n_{\text{eff1}} \cdot ff + n_{\text{eff2}} \cdot (1 - ff) \quad (3.1)$$

where ff denotes the fill factor of the grating coupler. With the effective index of refraction of the grating, we can get the grating period, Λ , from the Bragg condition:

$$n_c \cdot \sin\theta = n_{\text{eff}} - \frac{\lambda}{\Lambda} \quad (3.2)$$

where n_c denotes the effective index of the fibre mode, θ denotes the incident angle, n_{eff} denotes the effective index of refraction of the grating, and λ denotes the desired central wavelength. In our case, the grating period was

3.1. Detuned Shallow Etched Grating Coupler

calculated to be 660 nanometers. The insertion loss and back reflection of the initial grating coupler are shown in Fig. 3.1. The central wavelength of the initial grating coupler design is 1553nm, with an insertion loss of -3.1dB and a 3dB bandwidth of 70nm.

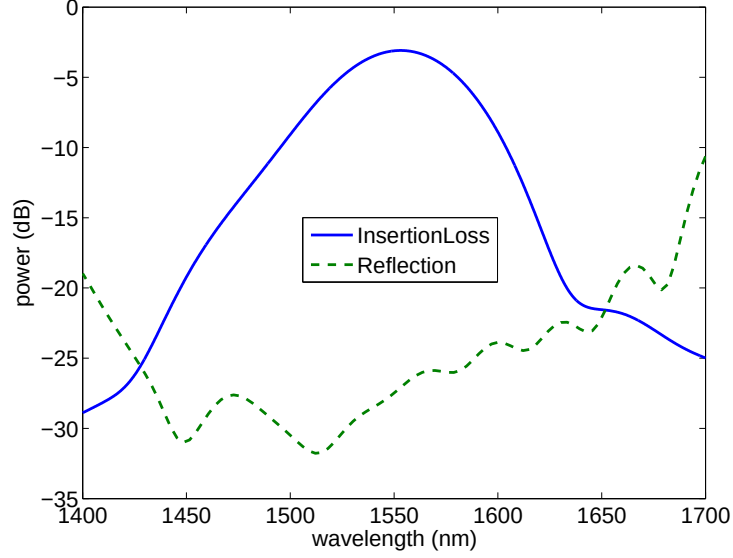


Figure 3.1: Insertion loss and back reflection to the waveguide of the initial grating coupler

3.1.2 Design Parameters

Once we got the initial condition, optimization of the design can be achieved by sweeping different design parameters such as the grating period, the duty cycle, the etch depth, and the incident angle. The impacts that each of the parameters has on the grating coupler are described in the following sections.

Grating Period

Grating period influences the performance of a grating coupler through the following equation:

$$\Lambda = \frac{\lambda}{n_{\text{eff}} - n_c \cdot \sin(\theta)} \quad (3.3)$$

3.1. Detuned Shallow Etched Grating Coupler

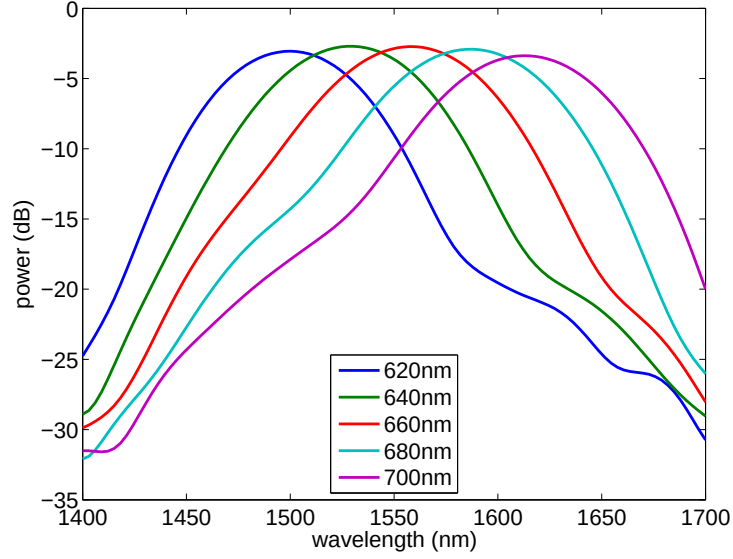


Figure 3.2: Variations of grating period

where Λ denotes the period of the grating coupler, λ denotes the central wavelength, n_{eff} denotes the effective index of the grating, n_c denotes the effective refractive index of the fibre mode and θ denotes the incident angle. As shown in the equation, the central wavelength of the grating coupler is proportional to the grating period. Figure. 3.2 shows the simulation results for varying the grating period. We kept $n_c = 1$, $\theta = 20^\circ$, and $ff = 0.5$. As we varied the grating period from 620 nanometers to 700 nanometers, the central wavelength of the grating coupler shifted from 1500 nanometers to 1612 nanometers. The redshift shown in the central wavelength is consistent with our analytical calculation. And the tuning coefficient of the grating period, which is defined as $\delta\lambda/\delta\Lambda$, was calculated to be 1.4 nm/nm .

Duty Cycle

Duty cycle affects the performance of a grating coupler through its impact on the effective index of refraction of the grating:

$$n_{\text{eff}} = n_c \cdot \sin(\theta) + \frac{\lambda}{\Lambda} \quad (3.4)$$

where n_{eff} denotes the effective index of refraction of the grating, θ is the incident angle and λ is the central wavelength. For a given grating period, the effective index of refraction of the grating is proportional to the duty cycle of the grating. Figure. 3.3 shows the simulation results for varying duty cycle. We kept grating period constant at 660 nanometers and varied the duty cycle from 230 nanometers to 430 nanometers. The central wavelength shifted from 1536 nanometers to 1579 nanometers as we varied the duty cycle. The tuning coefficient of the duty cycle, which is defined as $\frac{\delta\lambda}{\delta W}$, was calculated to be 0.215 nm/nm . By comparing Fig. 3.3 and Fig. 3.2, we note that grating period has a stronger impact on the central wavelength of the grating than the duty cycle does.

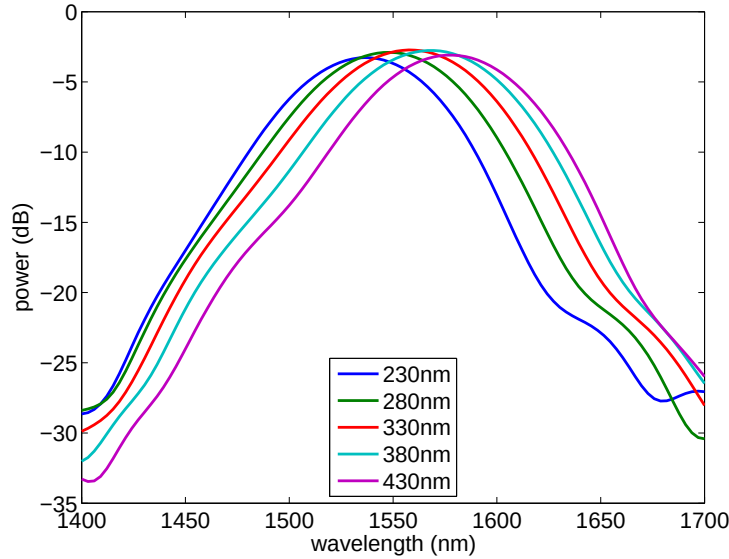


Figure 3.3: Variations of duty cycle

Etch Depth

Etch depth of a grating coupler also influences the performance of the grating coupler through its impact on the effective index of refraction of the grating:

$$n_{\text{eff}} = n_c \cdot \sin(\theta) + \frac{\lambda}{\Lambda} \quad (3.5)$$

where n_{eff} denotes the effective index of refraction of the grating, θ denotes the incident angle, and λ denotes the central wavelength. As the etch depth increases, the effective index of refraction of the shallow etched area decreases, thus n_{eff} decreases. The effective index of refraction is proportional to the central wavelength of the grating, so the etch depth of the grating coupler is inversely proportional to the central wavelength of the grating. Figure. 3.4 shows the simulation results for varying the etch depth. We

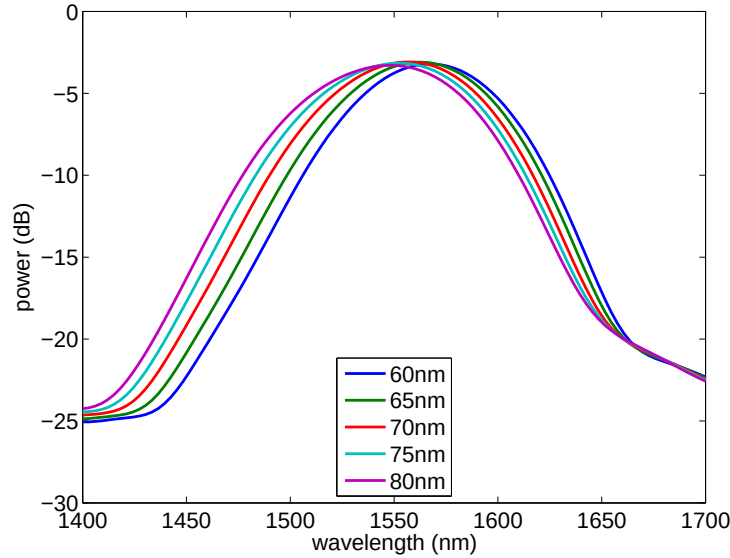


Figure 3.4: Variations of etch depth

kept grating period, duty cycle, incident angle constant and varied the etch depth of the grating coupler from 60 nanometers to 80 nanometers. The central wavelength shows a blueshift as the etch depth increases, which is

3.1. Detuned Shallow Etched Grating Coupler

consistent with our analytical calculation. The tuning coefficient of the etch depth, which is defines as $\delta\lambda/\delta ed$, was calculated to be $1.9 \text{ nm}/\text{nm}$.

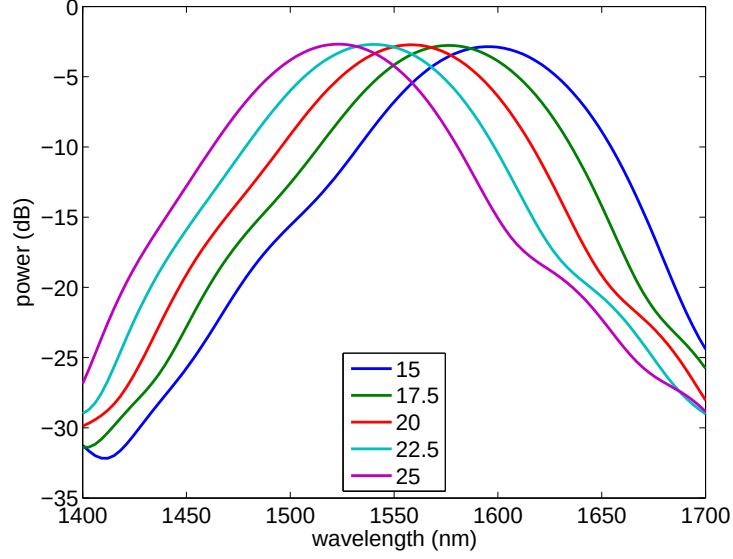


Figure 3.5: Variations of incident angle

Incident Angle

The incident angle of a grating coupler is defined as the angle between the incident wave (or out-coupled wave) and the normal to the grating surface. A positive angle indicates the case in which the incident wave and the coupled wave in the waveguide propagate in the same direction and a negative angle indicates the case in which the incident wave and the coupled wave in the waveguide propagate in opposite directions. The incident angle influences the central wavelength of the grating coupler through the following equation:

$$\sin(\theta) = \frac{n_{\text{eff}} - \frac{\lambda}{\Lambda}}{n_c} \quad (3.6)$$

where n_{eff} denotes the effective index of refraction of the grating, λ is the central wavelength, Λ is the grating period and n_c denotes the effective index

3.1. Detuned Shallow Etched Grating Coupler

of the wave incident on the grating. Simulation results for varying incident angle are shown in Figure 3.5. We kept the grating period, duty cycle and etch depth of the grating constant. As we increased the incident angle from 15 degree to 25 degree, the central wavelength shifted from 1594 nanometers to 1524 nanometers. So the tuning coefficient, which defined as $\delta\lambda/\delta\theta$, was calculated to be 7 nm/degree .

From the simulation results shown above, we can see that grating period, duty cycle, etch depth and incident angle all have impacts on the central wavelength of the grating coupler, but the tuning coefficient is different. Table 3.2 shows the comparison of the tuning coefficients of various parameters.

Parameter	period	duty cycle	etch depth	incident angle
Tuning coefficient	1.4 nm/nm	0.215 nm/nm	1.9 nm/nm	7 nm/degree

Table 3.2: Tuning coefficients of various parameters

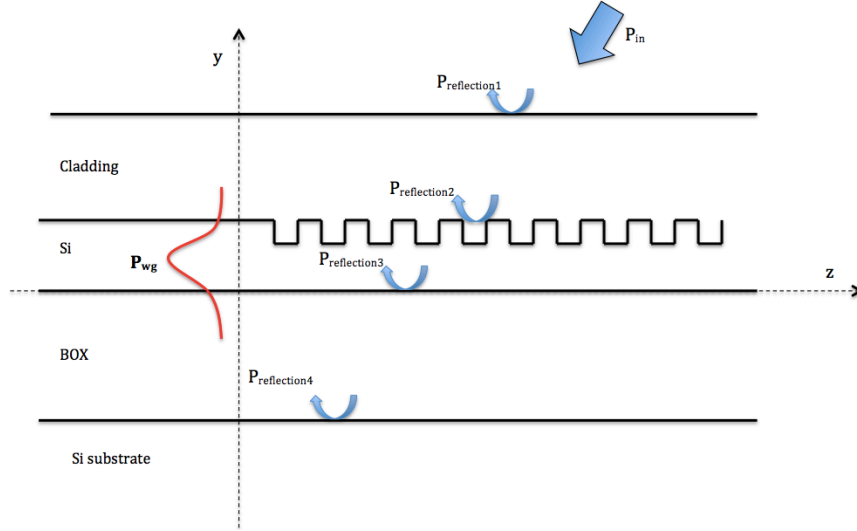


Figure 3.6: Schematic of reflections at different interfaces

Cladding and Buried Oxide

The thickness of the buried oxide and the thickness of the cladding are two important factors that have impacts on the insertion loss of the grating coupler. An illustration of different reflections at various interfaces of the grating coupler is shown in Figure. 3.6. The phase conditions between those reflections determine the insertion loss of the grating coupler. Minimum insertion loss can be achieved when $P_{\text{reflection1}}$ and $P_{\text{reflection2}}$ result in destructive interference and $P_{\text{reflection3}}$ and $P_{\text{reflection4}}$ result in destructive interference.

Simulation results for varying buried oxide are shown in Figure. 3.7. As we varied the thickness of the buried oxide from 1 μm to 3 μm , the insertion loss of the grating coupler changed in a sinusoidal way, which is determined by the phase condition between $P_{\text{reflection1}}$ and $P_{\text{reflection2}}$. The thickness of the buried oxide for a particular wafer type is chosen to achieve constructive interference between $P_{\text{reflection1}}$ and $P_{\text{reflection2}}$, therefore, low insertion loss can be obtained.

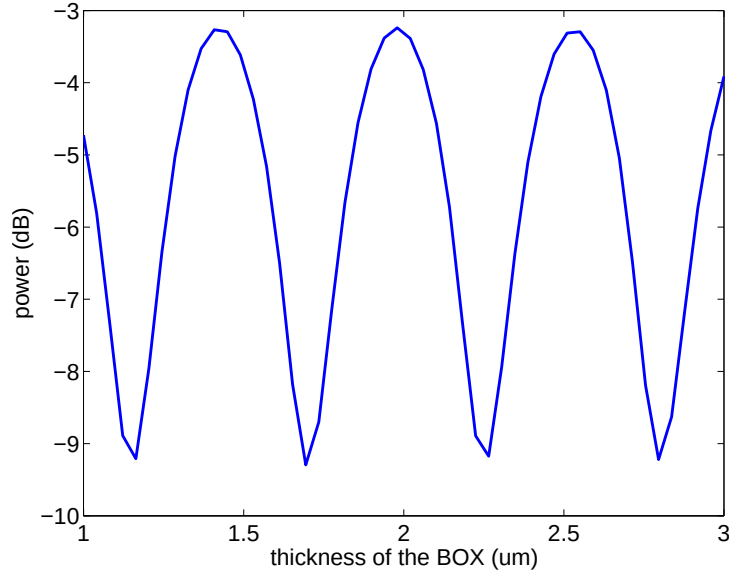


Figure 3.7: Variations for BOX thickness

3.1. Detuned Shallow Etched Grating Coupler

Similarly, the phase condition of $P_{\text{reflection3}}$ and $P_{\text{reflection4}}$ varies as the thickness of the cladding changes. We define the thickness of the cladding to be the height from the interface of the silicon and buried oxide to the top surface of the cladding. Figure. 3.8 shows the simulation results for varying the cladding thickness. Minimum insertion loss achieved where destructive interference occurs and maximum insertion loss achieved where constructive interference occurs. Depending on the incident angle and the central wavelength, the optimal thickness of cladding changes. By comparing Fig. 3.7 and Fig. 3.8 we note that the thickness of the buried oxide has a larger impact on the insertion loss of the grating. This is the case because the reflection coefficients of $P_{\text{reflection3}}$ and $P_{\text{reflection4}}$ are larger than those of $P_{\text{reflection1}}$ and $P_{\text{reflection2}}$.

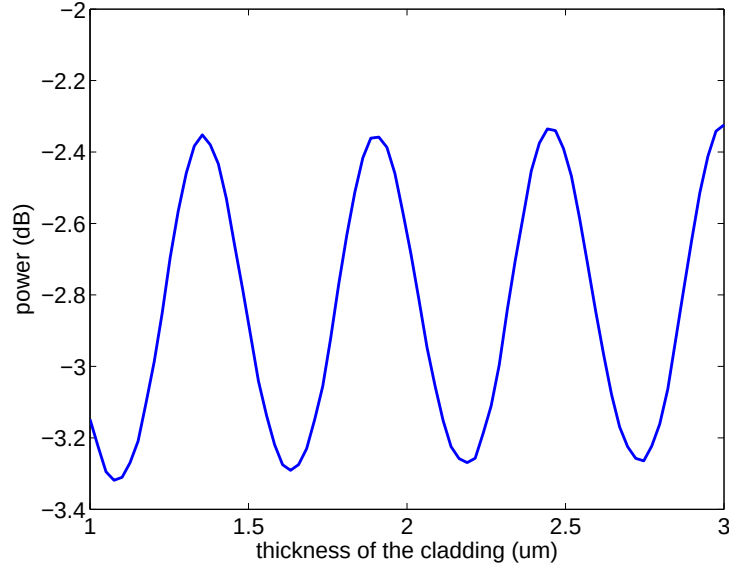


Figure 3.8: Variations for cladding thickness

3.1.3 Optimization of the Grating Coupler

Optimization of a grating coupler involves simulation sweeps of various parameters such as grating period, duty cycle, and incident angle, etc, but

3.1. Detuned Shallow Etched Grating Coupler

not all of parameters are variable. The thickness of different layers are determined by the wafer type, whereas the etch depth and the cladding are normally determined by the fabrication process. So the design variables are the grating period, duty cycle, and incident angle. The optimization process requires a lot of simulation sweeps on those design variables. Figure. 3.9 shows the simulation and measurement results of the grating coupler designed for the fabrication process provided by Opsis-IME [1]. The designed grating coupler has a grating period of 650 nanometers with a duty cycle of 350 nanometers. The simulated results show an insertion loss of -2.74dB with a 3dB bandwidth of 79.8nm, and the measurement shows an insertion loss of -4.64dB with a 3dB bandwidth of 74.9nm. The simulation results shown in Fig. 3.9 is obtained from the model shown in Fig. 2.9 (a) with the assumption that the distance between the fibre tip and the chip is negligible.

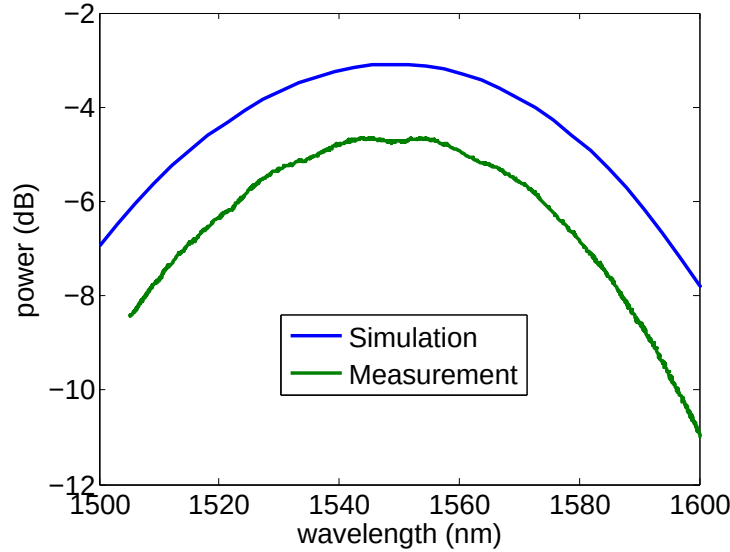


Figure 3.9: Grating coupler design fabricated through OpSIS-IME

From Fig. 3.9 we note that the measured insertion loss is smaller than the simulated insertion loss. In addition, the bandwidth of the measured spectrum is narrower than that of the simulated one. The mismatches in insertion loss and bandwidth mainly results from the gap between the fibre

3.1. Detuned Shallow Etched Grating Coupler

ribbon and the photonics chip. The images of the fibre ribbon used in our measurement setup are shown in Fig. 3.10. The tip of the fibre ribbon is polished at 20.3 degrees. According to Snell's Law, if the fibre tip is parallel to the photonics chip, it will give us a 30 degree incident angle in the air. In order to get a 20 degree incident angle in air, we need to rotate the fibre ribbon (as shown in the right image of Fig. 3.10). In this case, there will be a gap between the tip of the fibre ribbon and the photonic chip. In addition, extra space is left intentionally to protect the chip from scratching by the ribbon. So a large air gap exists between the ribbon and the photonic chip. The mode profile becomes larger as it propagates between the fibre ribbon and the chip, which introduces extra loss and narrows the bandwidth.

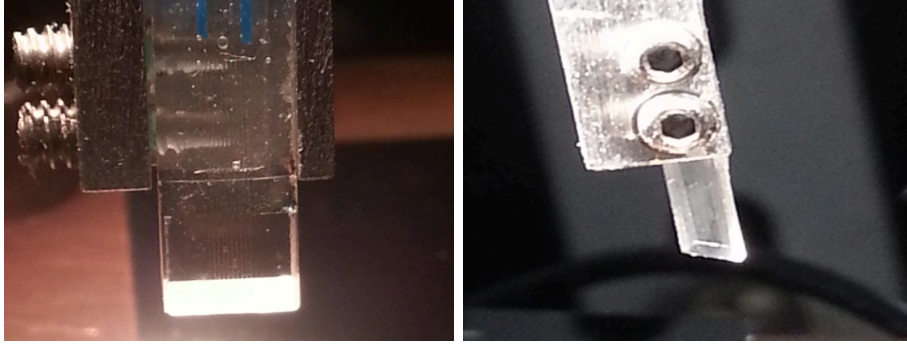


Figure 3.10: Images of fibre ribbon

Figure. 3.11 shows the impact of the gap between the fibre ribbon and the measured photonic chip. The blue curve indicates the change in insertion loss and the green curve indicates the change in bandwidth. For each value of the gap, i.e., the distance from the fibre to the chip, we optimized the XY position of the fibre to achieve lowest insertional loss. As we varied the gap from 0 to 200 nanometers, the insertion loss of the grating coupler decrease from -2.47dB to -6.08 dB and the bandwidth decreased from 77 nanometers to 44.5 nanometers.

3.1. Detuned Shallow Etched Grating Coupler

According to the simulation results shown in Fig. 3.11, we revised our simulation model and simulated the grating coupler design shown in Fig. 3.9 with a 25 μm air gap between the fibre ribbon tip and the chip. The comparison of the simulation results and measurement results of the grating coupler are shown in Fig. 3.12. The blue curve denotes the simulation results of the grating structure with 1 μm air gap between the fibre ribbon tip and the chip, the green curve denotes the simulation results with a 25 μm air gap between the grating and the chip, and the red curve denotes the measurement results for the same gating design. As we take the air gap between the fibre ribbon tip and the grating into consideration, a closer match between the simulation results and the measurement results have been achieved. The remaining mismatch of insertion loss mainly comes from the optical connections of the measurement system, such as the connections between the fibre and the laser and the connection between the fibre and the detector.

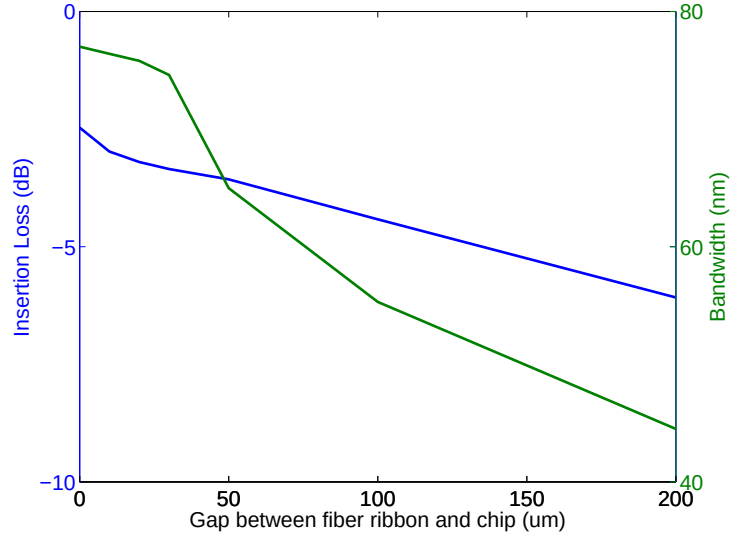


Figure 3.11: Impacts of the gap between fibre ribbon tip and photonic chip on the insertion loss and bandwidth of the grating coupler

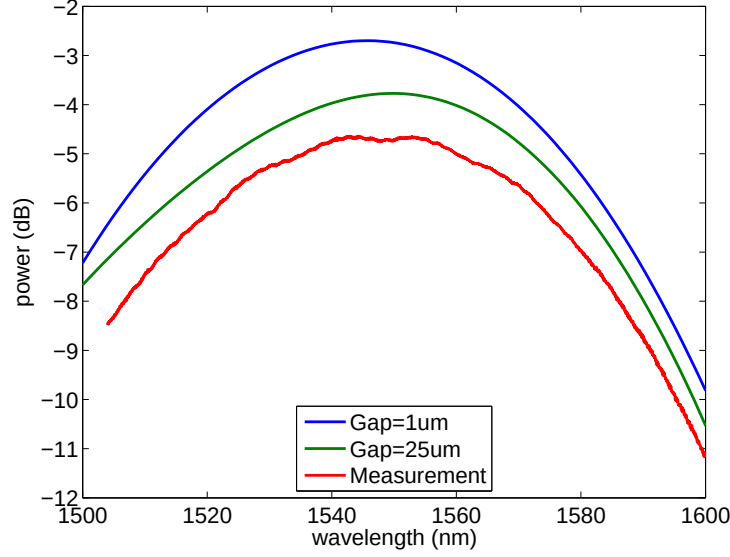


Figure 3.12: Spectra of simulation results with different gap distance and measurement results

3.1.4 Design Stability

To examine the stability of the grating coupler design at different positions on the same photonic chip, we measured the performance of the same grating coupler design at different positions of the chip. The fabrication of the grating couplers were done by electron beam lithography at University of Washington. The grating coupler is designed based on a wafer with $2\mu\text{m}$ buried oxide and 220 nm top *Si* layer. A shallow etch layer with an etch depth of 70 nm was used and air was employed as the cladding. The insertion losses, 3dB bandwidths, and the central wavelengths of the same design at 10 different positions were measured and the comparison of these results are shown in Fig. 3.13, Fig. 3.14, and Fig. 3.15. Fig. 3.13 shows the insertion losses of the same grating coupler design at different positions of the chip. The measured insertion losses are between -7.58dB and -8.24 dB . In order to protect the chip from scratching, we intentional left a gap between the chip the fibre tip. Therefore, the insertion loss of the measurement results is

3.1. Detuned Shallow Etched Grating Coupler

much larger than the simulation results. However, the stability in insertion loss of the grating coupler has been observed. Fig. 3.14 shows the measurement results of 3dB bandwidths of the same grating coupler at 10 different positions. The measured 3dB bandwidths of the grating coupler is between 47.5 nm and 42 nm . The chip may not perfectly aligned horizontally, which results in different gap distances between the fibre ribbon tip and the chip at different positions. The variations in gap distances is the main source for differences in bandwidth. Fig. 3.15 shows the central wavelengths of the same grating coupler design at different positions of the chip. Most of the measured grating couplers have a central wavelength of about 1550 nm , which is the designed central wavelength. The small discrepancy in central wavelength indicates the high accuracy of the fabrication process.

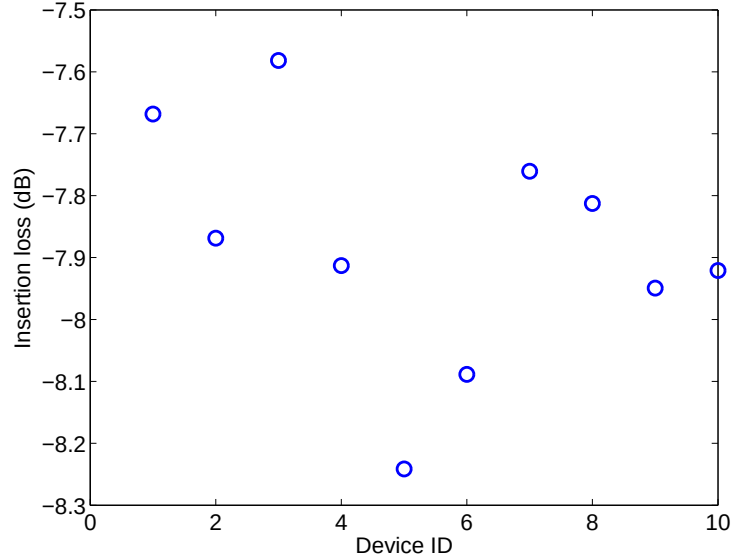


Figure 3.13: The insertion losses of the same grating coupler design at different positions of the chip

3.1. Detuned Shallow Etched Grating Coupler

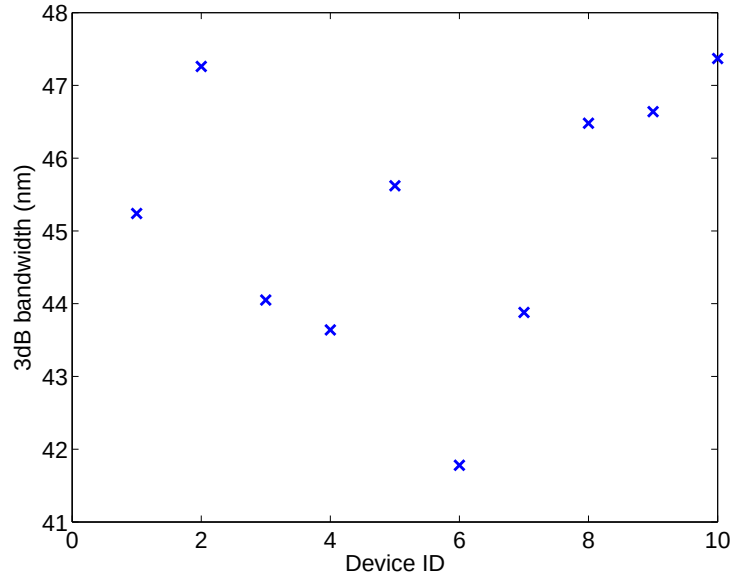


Figure 3.14: The 3dB bandwidths of the same grating coupler design at different positions of the chip

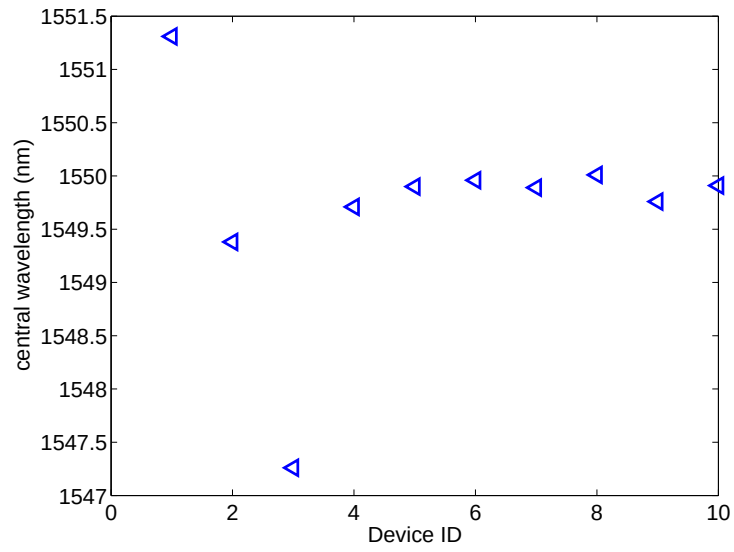


Figure 3.15: The central wavelengths of the same grating coupler design at different positions of the chip

3.2 Universal Grating Coupler Design

Depending on the fabrication process and applications, various grating couplers are required [16, 32, 35, 44, 56]. The traditional way of designing grating couplers involves brute-force simulations on all of the design parameters, which is very time-consuming and a lot of computational memory is required. Here we present a methodology for designing grating couplers based on theoretical calculations instead of numeric simulations. This methodology is enabled by the combination of Bragg condition and Effective Index Method (EIM). It has been validated for the wavelengths from 1260 *nm* to 1675 *nm* for both TE and TM mode waves. Based on this method, we also generated a script to draw the mask layout for grating couplers having only the central wavelength and the incident angle as inputs. The script accommodates various etch depths, silicon thickness (e.g., 220nm, 300nm), and cladding material (e.g., silicon oxide or air). This methodology has been verified by both FDTD simulations and measurement results.

3.2.1 Design Approach

The design flow of the proposed design method is shown in the flow chart on the next page. For any specific grating coupler there are two types of input parameters: process determined parameters and design intent parameters. The process determined parameters include the etch depth, the cladding material, and the thickness of each layer, which are determined by the fabrication process and wafer type. The design intent parameters include central wavelength, λ , incident angle, θ , and the polarization of the operational wave.

3.2. Universal Grating Coupler Design

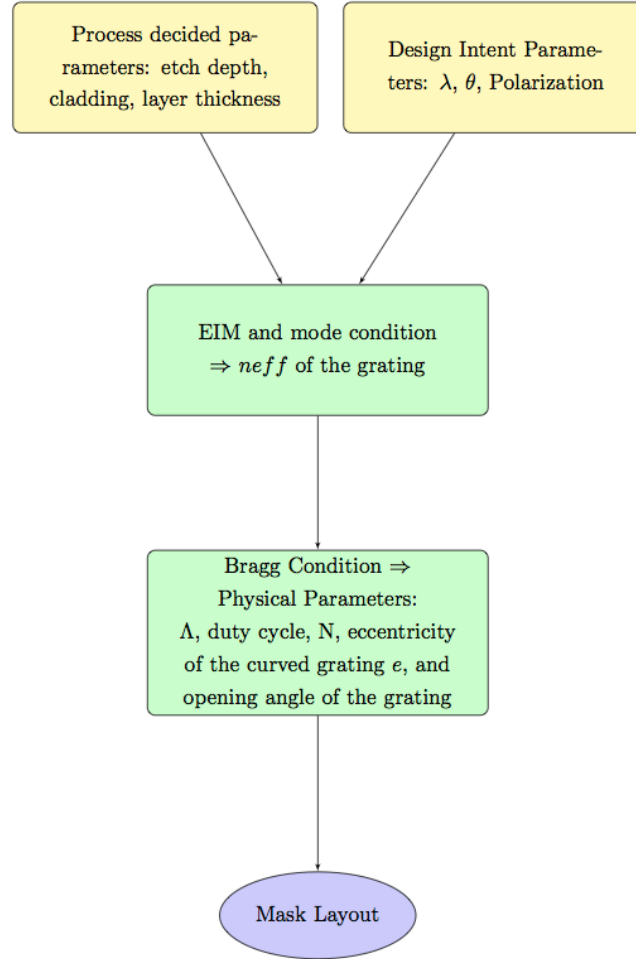


Figure 3.16: Flow chart of the universal design method

3.2. Universal Grating Coupler Design

Knowing the cladding material, the thickness of the top silicon layer, and the etch depth, we can obtain the effective index of the grating teeth and the effective index of the grating slots using the EIM, under the assumption that the grating has a infinite width. This assumption holds under the condition that grating couplers for an optical fibre normally have widths of more than 12 microns, which is much larger than the central wavelength. If we denote the effective index of the grating tooth to be n_{eff1} , and the effective index of refraction of the grating slots to be n_{eff2} , then the effective index of the grating can be expressed as:

$$n_{\text{eff}} = ff \cdot n_{\text{eff1}} + (1 - ff) \cdot n_{\text{eff2}} \quad (3.7)$$

where ff denotes the fill factor of the grating, which is defined as the ratio of the duty cycle and the grating period, i.e., $ff = w/\Lambda$.

After obtaining the effective index n_{eff} of the grating, the Bragg condition is employed to calculate the grating period:

$$n_c \cdot \sin\theta = n_{\text{eff}} - m \cdot \frac{\lambda}{\Lambda} \quad (3.8)$$

where n_c denotes the effective index of the fibre mode in the cladding, θ is the incident angle, λ is the desired central wavelength, and m denotes the diffraction order. For a specific grating coupler n_c , λ , and θ are known. In addition, the effective index of the grating n_{eff} can be calculated by the EIM. Therefore, we can get the period of the grating from Equation 3.18. This method is used to design a one-dimensional coupler, which can be simulated by 2D simulations in FDTD.

3.2.2 Simulation Results

Simulation models have been built using FDTD Solutions (Scripts shown in Appendix A). Given the process determined parameters and design intent parameters, our model can generate the desired grating couplers and simulate them. The input parameters used in our case are listed in Table 3.3.

Extensive simulations have been done for the wavelengths ranging from

3.2. Universal Grating Coupler Design

Top <i>Si</i> thickness	Etch depth	Cladding	λ	θ	Polarization
220 nm	70nm	air/oxide	1550 nm	10 degree	TE/TM

Table 3.3: Input paramters

1260nm to 1675nm, which covers all of the six optical bands. Air cladding and SiO_2 cladding have been examined for both TE and TM waves to verify the accuracy of the universal design method. The input parameters used in our case are shown in Table 3.3. Figure 3.17 shows the simulation results of the grating couplers for TE light with a 10 degree incident angle and SiO_2 as the cladding. The x-axis denotes the designed central wavelength, λ . The left y-axis, indicated by the blue curve, denotes the insertion loss of the grating couplers generated by the model, and the right y-axis, indicating by the green curve, denotes the wavelength mismatches. The wavelength mismatch, $\delta\lambda$, is defined as the difference between the actual central wavelength, λ_{real} , obtained from FDTD simulations and the design intent wavelength, λ , i.e., $\delta\lambda = \lambda_{\text{real}} - \lambda$. The insertion loss of the grating couplers varies as the wavelength changes, which results from the phase condition changes between $P_{\text{reflection3}}$ and $P_{\text{reflection4}}$ as shown in Fig. 3.6. Constructive interferences is obtained between $P_{\text{reflection3}}$ and $P_{\text{reflection3}}$ around 1310nm and 1550nm, therefore, the insertion losses of the grating couplers around these wavelengths are smaller. However, destructive interference between $P_{\text{reflection3}}$ and $P_{\text{reflection4}}$ is obtained around 1400nm, therefore, the insertion loss of the grating couplers around these wavelengths are larger. For most wavelengths, the simulated central wavelengths are close to the designed central wavelengths, especially around the two most commonly used optical windows around 1310 nm and 1550 nm. The wavelength mismatch can be calibrated by adjusting the incident angle during the measurement or by compensating for the wavelength during the design stage. Given that the tuning coefficient is 7 nm/degree for TE light and 10 nm/degree for TM light, the wavelength mismatches are within 2 degrees in most cases.

Figure. 3.18 shows the simulation results of the grating couplers for TM light with a 10 degree incident angle and SiO_2 as the cladding. The wave-

3.2. Universal Grating Coupler Design

length mismatches of grating couplers for TM light are larger than that of the grating couplers for TE light at some wavelengths. This is the case because the TM modes are less confined within the waveguides, therefore, the central wavelengths of the grating couplers for TM light are more sensitive to refractive index change than the central wavelengths of the grating couplers for TE light.

For comparison purposes, we simulated the grating couplers generated by the universal design model with a 10 degree incident angle and employed air as the cladding. Figure 3.19 shows the simulation results of the grating couples for TE operation with a 10 degree incident angle and air as the cladding. Figure 3.20 shows the simulation results of the grating couplers for TM operation with a 10 degree incident angle and air as the cladding. The insertion loss of the grating couplers with SiO_2 as the cladding are smaller than the ones with air cladding. This is the case because the refractive index contrast between the air and the top Si layer is reduced by employing the cladding layer, therefore, more light can be coupled into the grating by choosing the right thickness for the cladding layer. Also, the cladding layer mitigates the refractive index change of the grating as the period and the duty cycle of the grating varies, which results in the central wavelength being less sensitive to the changes in the duty cycle and grating period.

3.2. Universal Grating Coupler Design

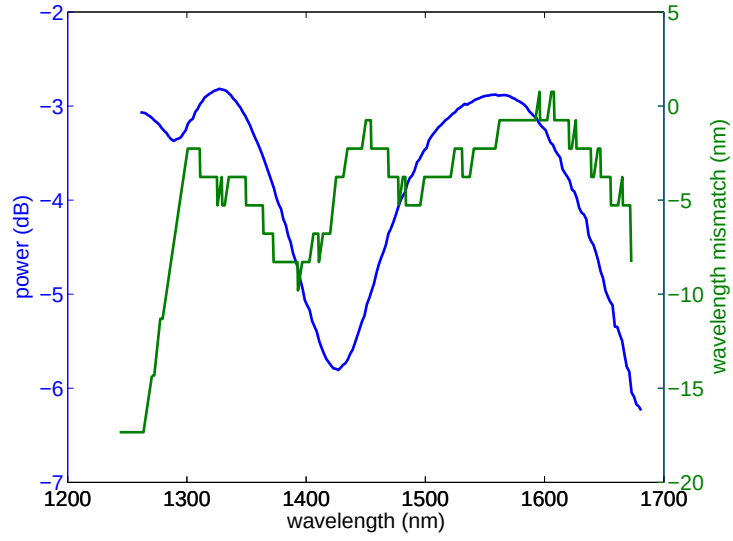


Figure 3.17: Universal grating couplers with 10 degree incident angle for TE mode wave with oxide cladding

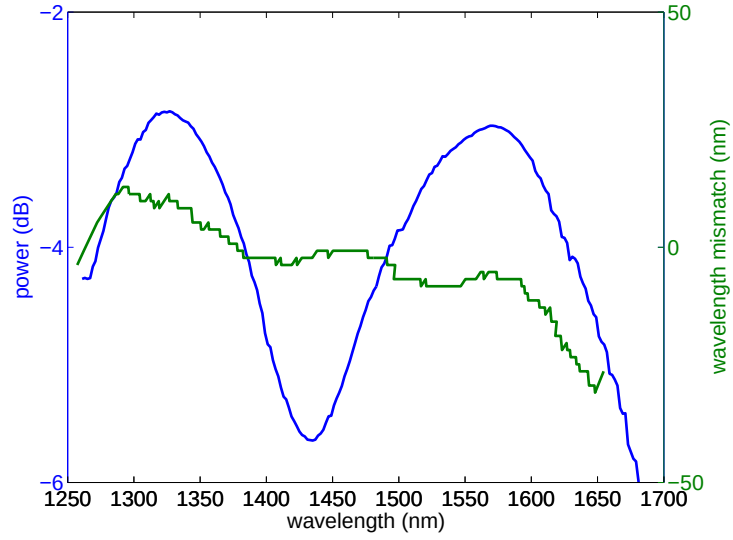


Figure 3.18: Universal grating couplers with 10 degree incident angle for TM mode wave with oxide cladding

3.2. Universal Grating Coupler Design

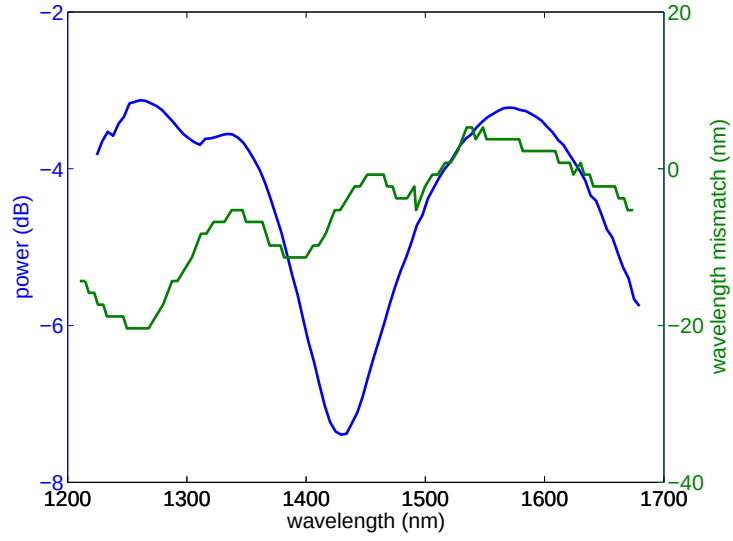


Figure 3.19: Universal grating coupler with 10 degree incident angle for TE mode wave with air cladding

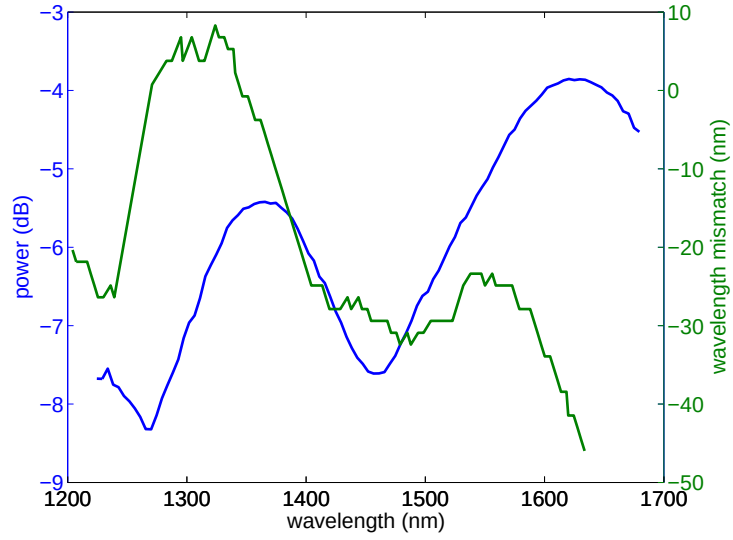


Figure 3.20: Universal grating coupler with 10 degree incident angle for TM mode wave with air cladding

3.2. Universal Grating Coupler Design

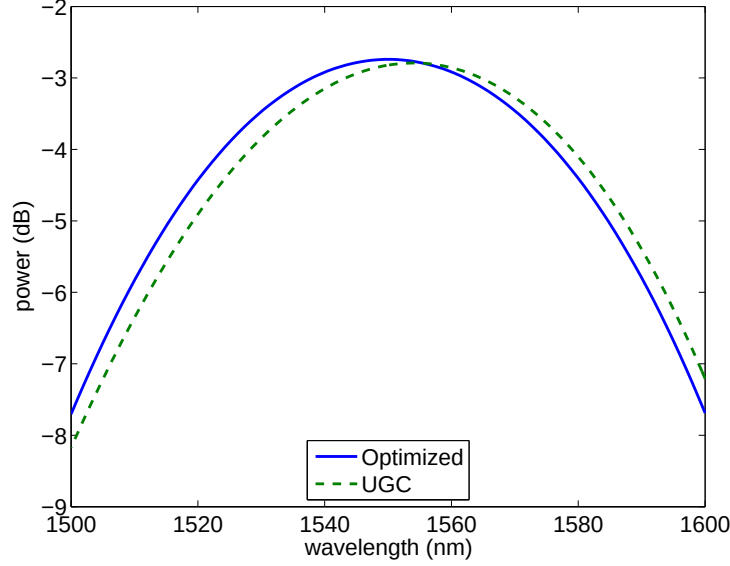


Figure 3.21: Comparison of designs generated by universal grating coupler model and the optimized design for 1550nm TE wave with 10 degree incident angle

In order to see the accuracy of the universal grating coupler design method, we picked the most commonly used wavelength in telecommunications, i.e., 1550 nm, to compare the spectrum of the grating coupler generated by the universal design model and the spectrum of the grating coupler design optimized by FDTD simulation. Figure. 3.21 shows the comparison of the simulation results of the grating coupler generated by the universal design method and the grating coupler optimized by FDTD simulations. The green dash curve denotes the spectrum of the grating coupler generated by the universal design method, which has an insertion loss of -2.79 and a 3dB bandwidth of 78.8 nm, and the blue curve denotes the spectrum of the grating coupler design optimized by FDTD simulations, which has an insertion loss of -2.74 and a 3dB bandwidth of 79.8 nm. The central wavelength mismatch between the two grating couplers is as small as 3 nanometers and the insertion loss only improved by 1% after hundreds of simulation sweeps for different design variables. In this thesis, the optimization of the grat-

ing coupler we use only considered uniform gratings. The improvements in the insertion loss can be obtained by more sophisticated designs with non-uniform gratings [50].

3.2.3 Mask Layout

With the physical parameters calculated from the Bragg condition, we can generate the mask layout of the grating coupler. So far, we have only addressed the grating couplers with straight gratings. For those grating couplers, two dimensional tapers are required to convert the mode from the grating into the waveguide. Typically, the width of the grating is about 12 microns, and the width of the waveguide is only about 500 nanometers. To achieve lossless conversion between the grating and the waveguide, two dimensional tapers with a length on the order of a few hundred microns are required [56], which is not space efficient. To tackle this issue, curved grating designs have been employed to make the grating couple more compact [54]. According to [54], compact grating structure can be obtained by curving the grating lines following the equation:

$$q\lambda_0 = n_{\text{eff}}\sqrt{y^2 + z^2} - zn_t\cos(\theta_c) \quad (3.9)$$

where q is the integer indicating each grating lines, θ_c denotes the angle between the fibre and the chip surface, n_t denotes the refractive index of the environment, λ_0 denotes the central wavelength, and n_{eff} denotes the effective index of the grating. The expression of Bragg condition for the curved gratings can be also converted to a cylindrical system [32]:

$$n_e kr = n_c k r \sin(\theta) \cos(\phi) + 2\pi N \quad (3.10)$$

where n_e denotes the effective of the mode propagating inside the grating, k denotes the wave vector in free space, n_c denotes the effective index of the wave in the cladding, θ denotes the incident angle, and ϕ denotes the angle subtended by an arbitrary points on the grating and to the z-axis. We generated a universal script in Pyxis (shown in Appendix B), a commercial

3.2. Universal Grating Coupler Design

software from Mentor Graphics, to draw the mask layout of the desired grating couplers with curved gratings. Our script is based on the cylindrical expression of the curved grating as shown by Equation. 3.10. An example of the mask layout of the grating coupler generated by our model is shown in Fig. 3.22.

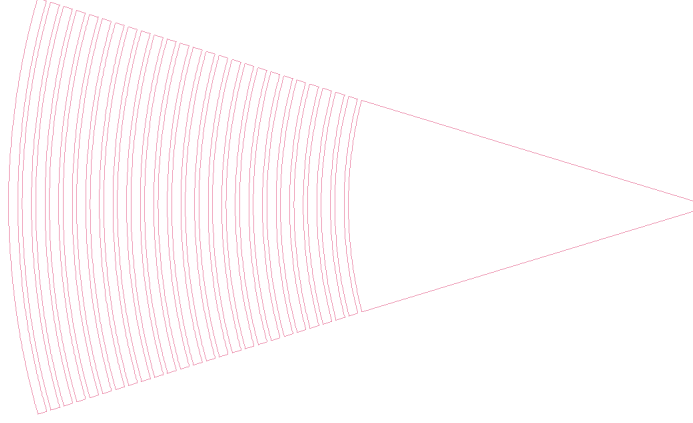


Figure 3.22: Mask layout of a grating coupler with focusing grating curve

3.2.4 Measurement Results

Test structures of the grating couplers generated by the universal design method have been fabricated through the electron beam lithography at University of Washington using the JEOL JBX-6300FS E-Beam Lithography system. Standard SOI wafer with a 220-nanometer top silicon layer and a 2-micron buried oxide were used for the fabrication. Silicon oxide was employed as the cladding layer to protect the functional structures. The etch depth of the shallow etched grating couplers is 70 nanometers. The grating couplers were designed for TE mode wave with incident angles of 10 degree, 15 degree and 20 degree. The target central wavelengths are 1500nm, 1520nm, 1540nm, 1560nm, 1580nm, and 1600nm.

Figure 3.23 shows the simulated and measured spectra of the grating couplers generated by the universal design method with a 10 degree incident

3.2. Universal Grating Coupler Design

angle. The simulation and measurement results of the grating couplers with the same central wavelength are indicated by the same colour. Due to the misalignment, inevitable system loss and fabrication errors, the measured insertion loss of the grating couplers are larger than the simulated results. In addition, the bandwidths of the measured grating coupler are narrower than the simulated ones, which is hypothesized to be originating from the large air gap. Similarly, Figure 3.24 shows the simulated and measured spectra of the grating couplers generated by the universal design method with a 15 degree incident angle, and Fig. 3.25 shows the simulated and measured spectra of the grating couplers generated by the universal design method with a 20 degree incident angle. The spectra of the measurement results follows the same trend as the simulation results but with the addition of extra insertion loss. The mismatch in central wavelength between the simulation results and measurement results may caused by the fabrication errors, e.g., the possible fabrication errors in grating period, duty cycle, and the thickness variation of the *Si* layer across the samples. The sensitivity of central wavelength of the design as function of design parameters such as grating period, duty cycle etc have been shown in Table 3.4, which has been explained in the previous section.

Parameters	period	duty cycle	etch depth	incident angle
Tuning coefficient	1.4	0.215	1.9	7 nm/degree

Table 3.4: Tuning coefficient of different parameters

3.2. Universal Grating Coupler Design

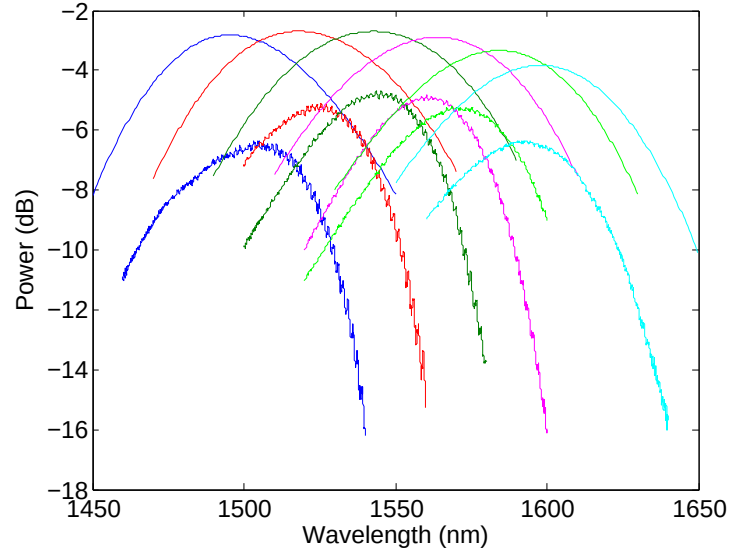


Figure 3.23: Measurement vs. simulation of universal grating couplers with 10 degree incident angle

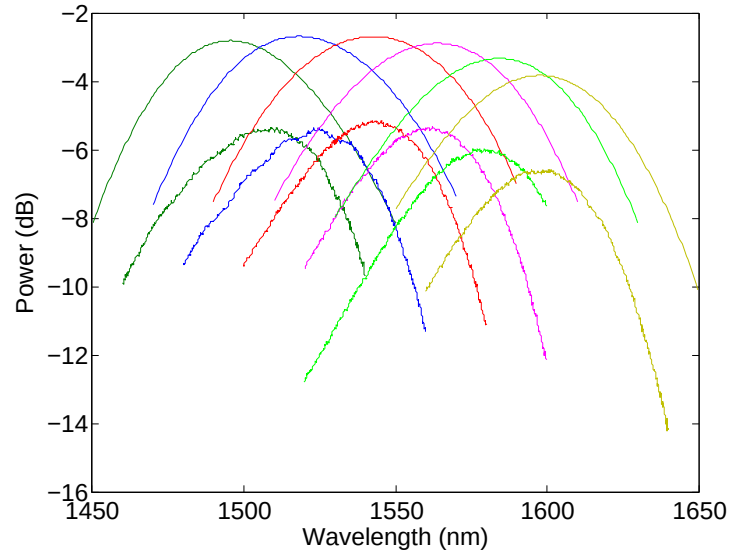


Figure 3.24: Measurement vs. simulation of universal grating couplers with 15 degree incident angle

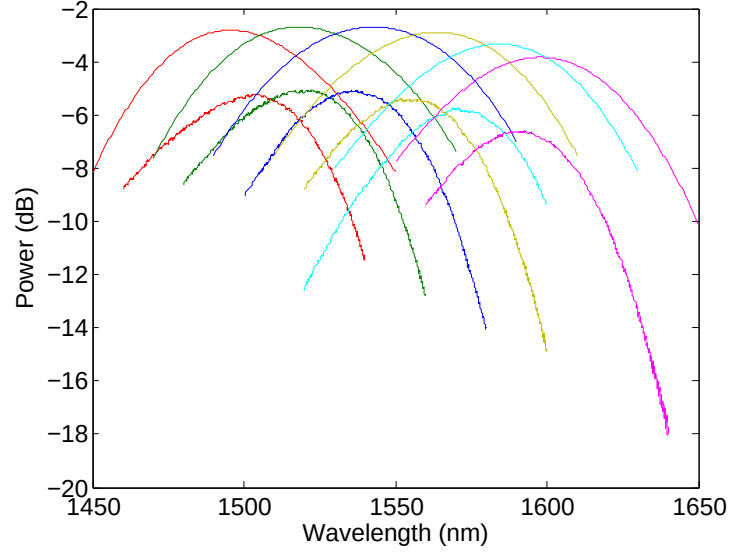


Figure 3.25: Measurement vs. simulation of universal grating couplers with 20 degree incident angle

Further comparison of simulation and measurement results of the grating couplers generated by the universal design method are shown in Fig. 3.26 to Fig. 3.28. Figure. 3.26 (a) shows the comparison of the simulated and measured insertion loss of the grating couplers with 10 degree incident angle. Figure. 3.26 (b) shows the comparison of the simulated and measured central wavelength mismatch. Similarly, the comparison between simulation and measurement results of the grating couplers with 15 degree incident angle and 20 degree incident angle are shown in Fig. 3.27 to Fig. 3.28. Obviously, the insertion loss of the simulation results and measurement results follows the same trends. However, the central wavelength mismatch of the measurement results shows a random behaviour, which is not consistent with the simulation results. The discrepancy in central wavelength is mainly caused by the fabrication error because the possible fabrication errors in grating period, duty cycle, etch depth all have impacts on the central wavelength of the grating coupler.

3.2. Universal Grating Coupler Design

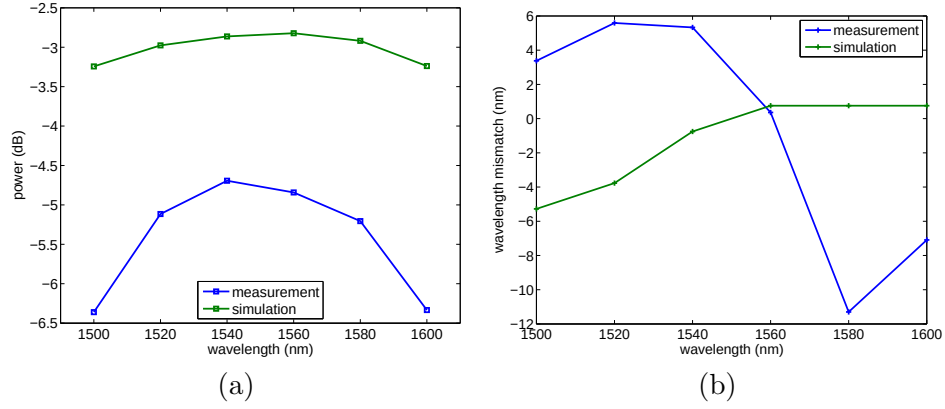


Figure 3.26: (a) Peak power of simulation and measurement results with 10 incident angle; (b) comparison of simulated and measured wavelength mismatch with 10 degree incident angle

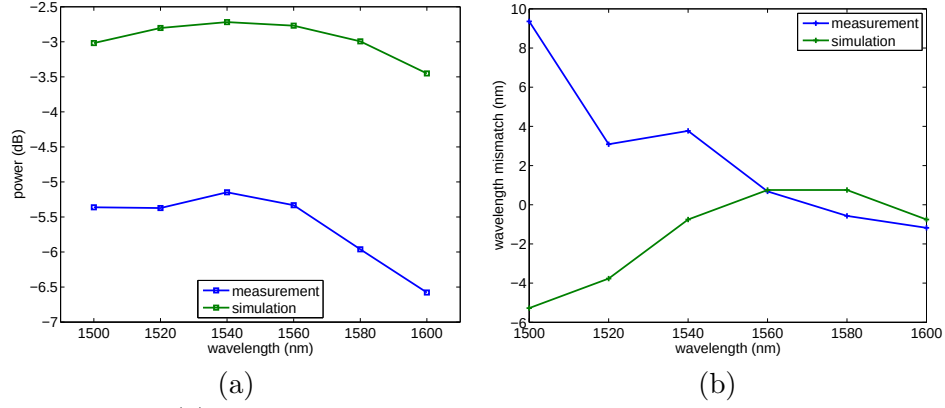


Figure 3.27: (a) Peak power of simulation and measurement results with 15 incident angle; (b) comparison of simulated and measured wavelength mismatch with 15 degree incident angle

3.2. Universal Grating Coupler Design

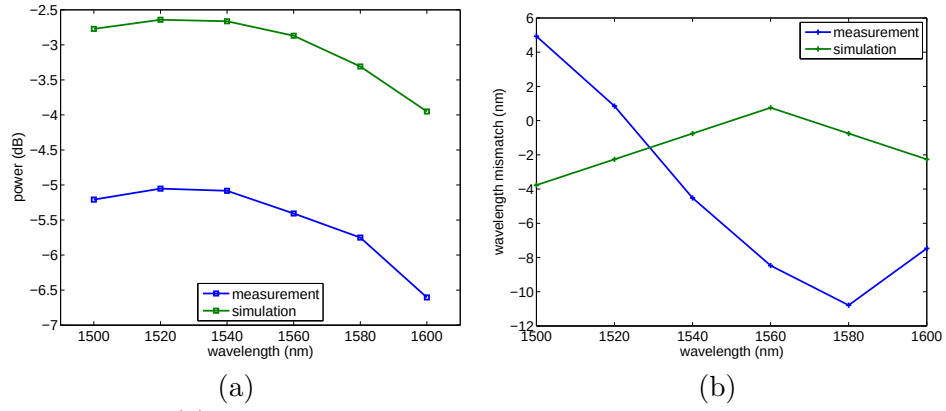


Figure 3.28: (a) Peak power of simulation and measurement results with 20 incident angle; (b) comparison of simulated and measured wavelength mismatch with 20 degree incident angle

Chapter 4

Fully Etched Grating Coupler

All of the grating couplers mentioned so far are shallow etched grating couplers, and the fabrication of these grating couplers requires two lithography steps. Shallow etched grating couplers have the advantage of smaller back reflection to the waveguide and lower insertion loss than the fully etched grating couplers. However, the additional lithography step also increases the fabrication cost and complexity. Fully etched grating couplers offer a fast and economic experimental solution for the coupling issue, especially for quick prototyping through electron beam lithography. All of the fundamental building blocks of the photonics integrated circuits can be fabricated in just one step by implementing the fully etched grating coupler. In this chapter, the performance of the traditional fully etched grating couplers will be presented first, and the drawbacks of which will be pointed out. Then, fully etched grating couplers with refractive index engineered sub-wavelength structures will be introduced to improve the performance of the fully etched grating couplers.

4.1 Regular Fully Etched Grating Coupler

Figure 4.1 is a schematic of a regular fully etched grating coupler in SOI with uniform gratings. The grating teeth are formed by fully etching through the functional silicon layer instead of partially etching. The thicknesses of the functional silicon layer and buried oxide used in our case are 2 μm and 220 nm , respectively. Air is employed as the cladding for the grating coupler design.

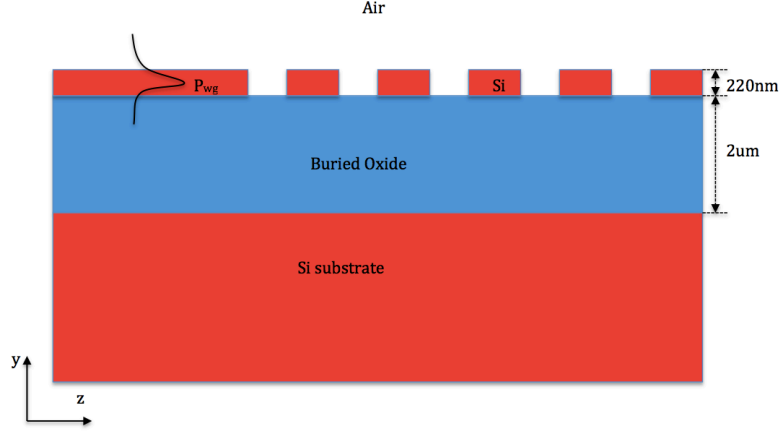


Figure 4.1: Schematic of fully etched grating coupler

4.1.1 Bottlenecks of Regular Fully Etched Grating Couplers

The disadvantages of the regular fully etched grating couplers include the poor insertion loss and the large back reflection to the waveguide. The poor insertion loss is mainly caused by the mode mismatch and penetration loss, and the large back reflection to the waveguide results from the large refractive index contrast between the waveguide and the grating.

When considering the grating coupler as an output coupler (shown in Fig. 4.1), the power in the waveguide will be exponentially decaying due to the presence of the grating [49]:

$$P_{wg}(z) = P_{wg}(z = 0) \exp(-2\alpha z) \quad (4.1)$$

where 2α is the coupling strength or leakage factor of the grating, $P_{wg}(z)$ denotes the power of the mode at z . The inverse of the coupling strength is defined as the coupling length, i.e., $L_c = (2\alpha)^{-1}$. For shallow etched grating couplers, the coupling strength α is small, and the coupling length L_c of the grating couplers is similar to the diameter of the mode from an optical fibre. Therefore, a large mode overlap between the exponential mode from the grating and the Gaussian mode of a fibre can be achieved. However,

4.1. Regular Fully Etched Grating Coupler

the coupling strength α of fully etched grating couplers is much larger than that of the shallow etched ones, therefore the coupling length L_c of the fully etched grating couplers is much smaller than that of the shallow etched ones, which results in a larger mode mismatch between the grating and the optical fibre.

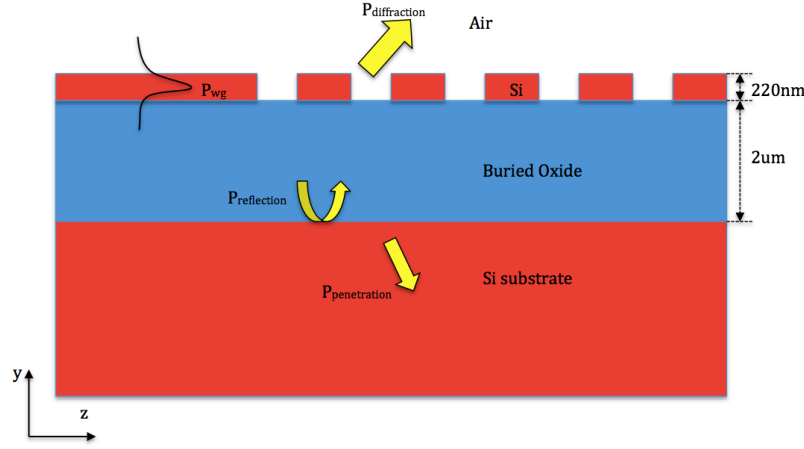


Figure 4.2: Schematic of diffraction, reflection and penetration of a regular fully etched grating coupler

The large penetration loss is another reason leading to the poor insertion loss of the fully etched grating couplers. Figure 4.2 shows the diffraction, reflection, and penetration behaviours of a fully etched grating coupler. As the incident wave impinges on the gratings, a portion of the light is diffracted upwards to the air and a portion of the light is scattered downward to the substrate. At the interface of the buried oxide and the Si substrate, part of the light scattered downward from the grating is reflected back to the grating and the other part of the light penetrates into the substrate. At the interface of the functional Si layer and the buried oxide, the reflected light from the lower interface interferes with the portion of light diffracted upwards. Depending on the phase condition, either constructive interference or destructive interference can be obtained. Lowest insertion loss can be achieved when constructive interference is obtained.

4.1. Regular Fully Etched Grating Coupler

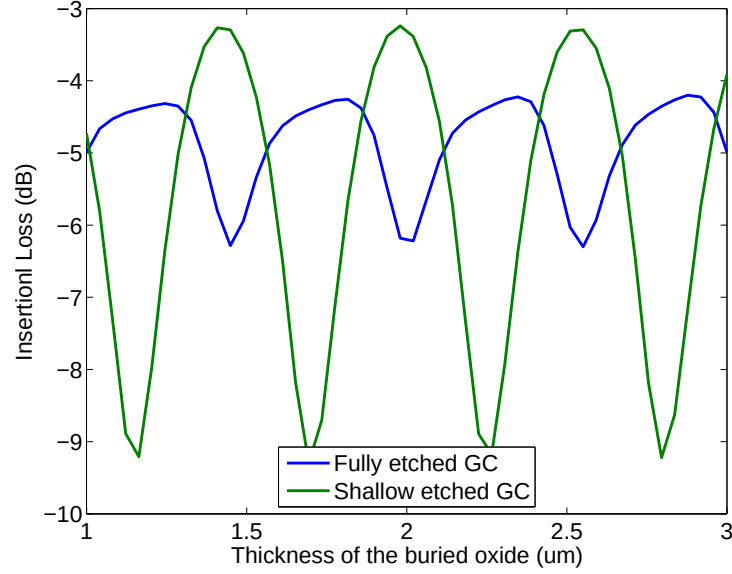


Figure 4.3: Insertion loss of grating couplers with 20 degree incident angle for TE operation wave as function of the thickness of the buried oxide.

The phase condition between the diffracted wave, i.e., $P_{\text{diffraction}}$, and the reflected wave, i.e., $P_{\text{reflected}}$, depends on the thickness of the buried oxide. Figure 4.3 shows the insertion loss of grating couplers with 20 degree incident angles for TE operation wave as function of the thickness of the buried oxide. As we vary the thickness of the buried oxide from 1 μm to 3 μm , the insertion loss of the grating couplers varies as the phase conditions changes. The standard SOI wafers normally have a 2- μm thick buried oxide, which is chosen for the shallow etched grating couplers, but not ideal for fully etched grating couplers.

The large back reflection to the waveguide of the fully etched grating couplers results from the Fresnel's reflection. The reflection coefficient for TE wave can be expressed as:

$$r^{TE} = \frac{n_1 \cdot \cos(\theta_i) - n_2 \cdot \cos(\theta_t)}{n_1 \cdot \cos(\theta_i) + n_2 \cdot \cos(\theta_t)} \quad (4.2)$$

where n_1, n_2 are the refractive indices of corresponding layers. And θ_i, θ_t are the incident angle and transmission angle, respectively. In our case, n_1 equals to the refractive index of silicon, i.e., $n_1 = 3.47$, and n_2 equals to the refractive index of cladding material. The Fresnel's reflection coefficient is as high as 0.414 in the case with silica as cladding material, which means more than 17% of the power will be reflected back into the waveguide. The back reflection to waveguide is even larger in the case with air as cladding since air has a lower refractive index than silica.

4.1.2 Optimization of the Regular Fully Etched Grating Couplers

Insertion loss, back reflection to the waveguide, and bandwidth are the most important figures of merit for a grating coupler. The optimization of a regular fully etched grating coupler involves optimizations for different design parameters such as incident angle, grating period, duty cycle, etc.

Two efficiencies determine the overall efficiency of the fully etched grating couplers. The first efficiency, η_1 , is defined as the ratio of the light scattered upwards and the input power from the waveguide:

$$\eta_1 = \frac{P_{\text{up}}}{P_{\text{wg}}} \quad (4.3)$$

where P_{up} denotes the portion of light scattered upwards and P_{wg} denotes the input power from the waveguide. η_1 is also called the directionality of the grating coupler. Theoretically, η_1 can be maximized by employing the optimal thickness of the buried oxide [11, 47, 54]. However, the thickness of the buried oxide is determined by the wafer type, which is not changeable. The second efficiency, η_2 , is defined as the ratio of the power coupled into the desired mode, i.e., the fundamental mode of the fibre, and the power scattered upward:

$$\eta_2 = \frac{P_{\text{fund}}}{P_{\text{up}}} \quad (4.4)$$

where P_{fund} denotes the power coupled in the fundamental mode of the fibre, and P_{up} denotes the power scattered upwards by the grating. This

4.1. Regular Fully Etched Grating Coupler

efficiency indicates the overlap between the exponential scattering mode from the grating and the Gaussian mode in an optical fibre. The exponential scattering pattern can be engineered by varying the coupling strength α of the grating coupler, and the maximized overlap was reported to be 80% [36], which is consistent with our simulation results shown in Fig. 4.4. The overall coupling efficiency (insertion loss) of the grating coupler can be expressed as the product of the two partial efficiency:

$$\eta = \eta_1 \cdot \eta_2 \quad (4.5)$$

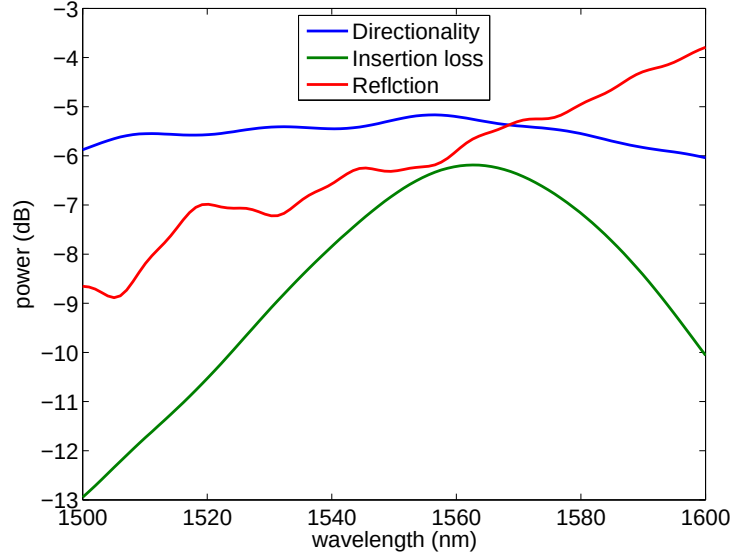


Figure 4.4: Directionality, insertion loss and reflection to waveguide of a general fully-etched grating coupler

An optimal design can be found by optimizing the two partial efficiencies η_1 and η_2 . The thickness of the buried oxide is determined by the wafer type, which is 2um in our case, η_1 is fixed with the given central wavelength and incident angle. The simulation results of a fully etched grating coupler with air cladding operated for TE mode at 1550nm have been shown in Fig. 4.4. The blue curve denotes the directionality of the grating, the green curve



Figure 4.5: Mask layout of a fully etched grating coupler test structure

denotes the insertion loss of the grating, and the red curve denotes the back reflection to the waveguide of the grating. Due to the large refractive index contrast between the waveguide and the grating, the back reflection of the fully etched grating coupler is huge, which is even larger than the insertion loss of the grating coupler.

A mask layout of a fully etched grating coupler test structure is shown in Fig. 4.5. An input coupler and an output coupler are connected by a u-shape silicon wire waveguide. The centre-to-centre distance between the two grating couplers is 127 microns, which is determined by the pitch of the fibre ribbon used in our lab. The comparison of simulation and measurement results for such a test structure is shown in Fig. 4.6. The green curve denotes the simulated insertion loss of the grating pair and the blue curve denotes the measured insertion loss of the same structure. Since the simulation result is obtained from the simulation of a single grating coupler, and the back reflection to the waveguide is not taken into consideration, no oscillations are shown in the simulation spectrum. Strong oscillations have been observed in the measurement spectrum of the grating structure, which is resulting from the large back reflection of the grating coupler. Such big ripples strongly affect the performance of resonator structures such as ring resonators, disks, and Bragg gratings, etc.

4.1. Regular Fully Etched Grating Coupler

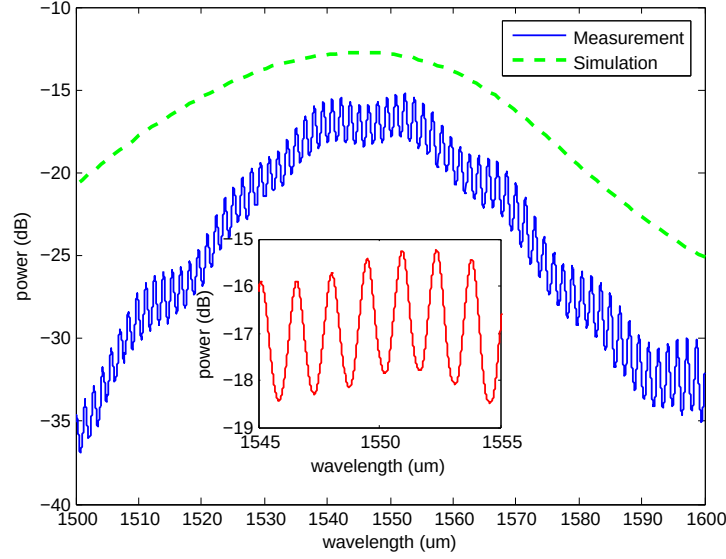


Figure 4.6: Comparison of measurement result and simulation result of fully-etched grating coupler

The extinction ratio of the ripples shown in the measurement spectrum can be calculated by the transmission function of the Fabry-Perot cavity:

$$T = \frac{(1 - R)^2}{(1 - R)^2 + 4 \cdot R \cdot \sin^2(\delta/2)} \quad (4.6)$$

where T denotes the transmitted power of a Fabry-Perot cavity, R is the reflectivity of the mirror, i.e., grating couplers in our case, and δ is the round trip phase shift. The reflectivity of the grating coupler is about 20% as shown in Fig. 4.4, so the extinction ratio of the ripples is about 2.8dB which is consistent with our measurement results shown in the inset of Fig. 4.6.

4.2 Improved Fully Etched Grating Coupler

As we have discussed in the previous section, the bottlenecks faced by the fully etched grating couplers are the large back reflection to the waveguide and the high insertion loss. The strong back reflection of the fully etched grating couplers results from the large refractive index contrast between the waveguide and the grating. The simulation results of the back reflections of both shallow etched grating couplers and fully etched grating couplers are shown in Fig. 4.7. The green curve denotes the normalized back reflection of a shallow etched grating coupler designed based on a SOI with a 2- μm buried oxide, 220 nm top Si layer, and air as the cladding. The blue curve denotes the back reflection of a fully etched grating coupler designed with the same wafer type and cladding. The grating couplers are designed for TE mode with a central wavelength of 1550 nm. The back reflection of the shallow etched grating coupler at the designed central wavelength is much smaller than that of the fully etched grating coupler and a difference of more than 20 dB has been observed.

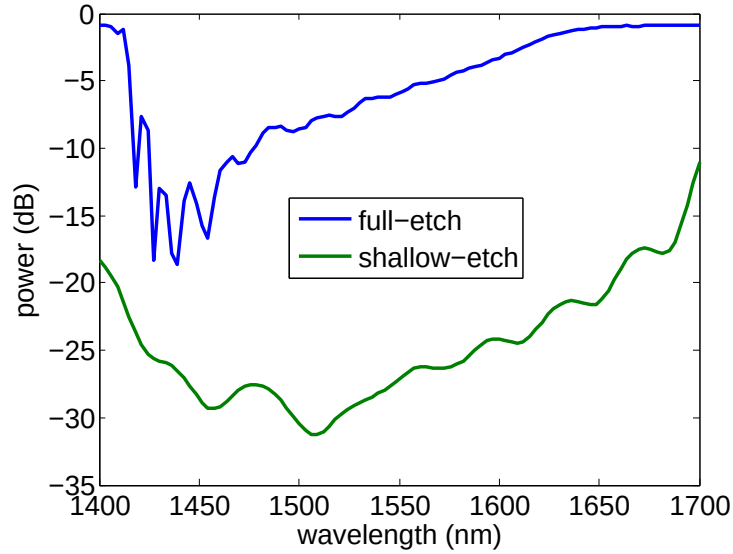


Figure 4.7: Comparison of back reflections between shallow-etched grating coupler and fully-etched grating coupler

Different approaches have been employed to tackle the issues of large back reflection and high insertion loss of fully etched grating couplers. Photonic crystal structures have been implemented in fully etched grating couplers to achieve low insertion loss [27]. However, customized wafers have been used to meet the requirement of the optimized design, which cannot be applied to the general cases with standard SOI wafers. Sub-wavelength structures [13, 14] have been used to reduce the back reflection and improve the insertion loss of the fully etched grating coupler at the same time. However, the reported designs are only applied for TM mode wave. In addition, all of the reported grating couplers are one-dimensional periodic structures and adiabatic tapers on the order of a few hundred micrometers are required to convert the optical mode from the grating to the waveguide, which waste a lot of precious space on the chip.

In this section, we will demonstrate a fully etched grating coupler for TE mode wave operated at 1550 nm with reduced back reflection and improved coupling efficiency. The proposed design is based on a SOI wafer with a $2\mu\text{m}$ buried oxide and 220 nm top *Si* layer. The proposed design also has the potential to be converted to a compact structure with curved gratings, which has a much smaller footprint.

4.2.1 Design Approach

In order to reduce the back reflection of the fully etched grating couplers, we proposed a fully etched grating with effective index areas consisting of air gaps and minor sub-wavelength gratings (show in Fig. 4.8). The major (wider) gratings determine the period and duty cycle of the grating coupler and the minor sub-wavelength gratings are employed to mitigate the refractive index contrast between the major gratings and the original air gaps. Those minor gratings are easier to be fabricated than the holes or rectangles [13, 14, 27]. Another advantage of the proposed design is the potential to be converted to compact structures with curved gratings. Based on the ideal mentioned in [6], one-dimensional gratings can be converted to more compact strcutures with curved gratings, which enables a much

smaller footprint.

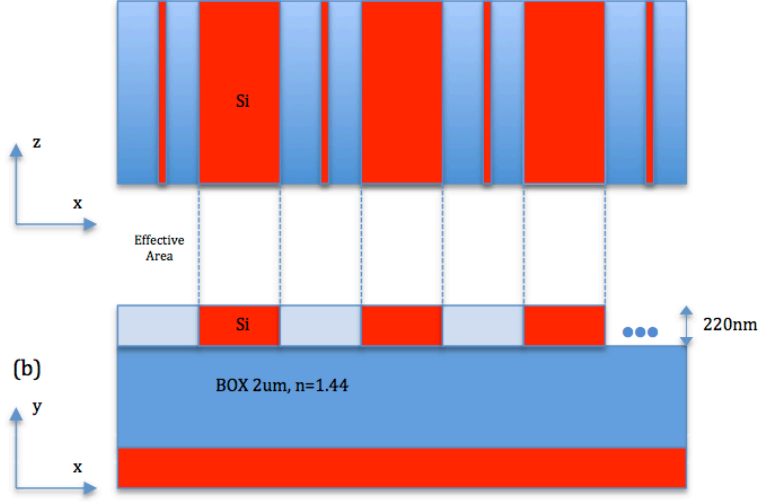


Figure 4.8: Schematic of a fully etched grating coupler with sub-wavelength gratings

According to the effective medium theory (EMT), zeroth-order approximation can be applied to sub-wavelength structures with period-to-wavelength ratio, defined as $R = n_{\text{eff}} \cdot \Lambda / \lambda$, much smaller than 1:

$$\frac{1}{n_{TE}^{(0)}} = \left[\frac{ff}{n_L^2} + \frac{(1-ff)}{n_{\text{si}}^2} \right]^{1/2} \quad (4.7)$$

where $n^{(0)TE}$ denotes the refractive index of the approximated grating region, n_L denotes the refractive index of the effective index area with minor sub-wavelength gratings, n_{si} denotes the refractive index of the major grating, and ff is defined as the fill factor of the grating. Second-order EMT can be used as a more accurate approximation to explore structures with lateral feature size of the same order as the wavelength in the medium [38]:

$$n_{TE}^{(2)} = n_{TE}^{(0)} \left[1 + \frac{\pi^2}{3} R^2 ff^2 (1-ff)^2 (n_L^2 - n_{\text{si}}^2)^2 \cdot \left(\frac{n_{TM}^{(0)}}{n_{\text{eff}}} \right)^2 \left(\frac{n_{TE}^{(0)}}{n_L n_{\text{si}}} \right)^4 \right]^{1/2} \quad (4.8)$$

4.2. Improved Fully Etched Grating Coupler

where n_{eff} denotes the effective index of the TE mode in the grating region.

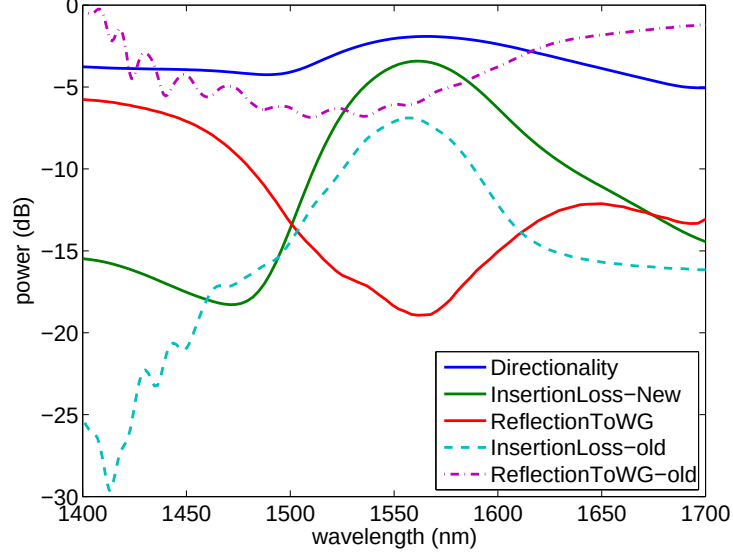


Figure 4.9: Comparison of regular fully etched grating coupler and the fully etched grating coupler with minor sub-wavelength gratings.

4.2.2 Simulation Results

FDTD method has been used to simulate and optimize the grating structure. The simulations were based on the SOI wafer we used for fabrication, with a $2\text{ }\mu\text{m}$ buried oxide and 220 nm top *Si* layer. Silicon oxide was employed as the cladding for protection. Particle Swarm Optimization (PSO), which is an optimization algorithm for electromagnetic optimization problems [39, 61], has been used to optimize the grating design. Design parameters, i.e., grating period, duty cycle, and the width of the minor gratings were optimized to achieve small back reflection as well as low insertion loss. The optimized design has a period of 625 nm , a duty cycle of 190 nm , and sub-wavelength gratings with a width of 50 nm .

Figure 4.9 shows the simulation results of a regular fully etched grating coupler and the fully etched grating coupler with minor sub-wavelength

gratings. The dash lines denote the insertion loss and the back reflection of the regular fully etched grating coupler, and the solid lines denote the directionality, insertion loss, and the back reflection of the proposed grating coupler. More than 64% (-1.9 dB) of the input power from the waveguide was diffracted upwards by the grating. However, due to the mode mismatch between the grating coupler and the fibre, only 45% (-3.47 dB) of light from the waveguide was coupled in the fundamental mode of fibre. The back reflection of the grating coupler is reduced from -6 dB to about -18 dB, which is similar to that of a shallow etched grating coupler.

4.2.3 Measurement Results

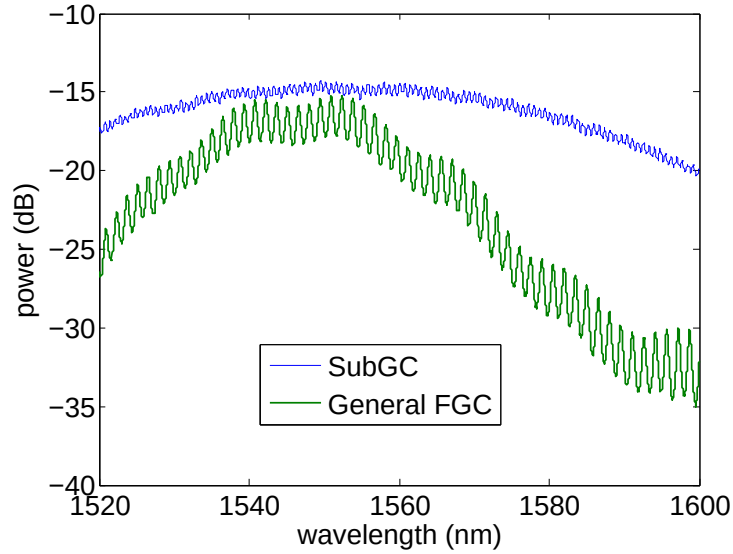


Figure 4.10: Measurement results of the regular fully etched grating coupler and the fully-etched grating coupler with minor sub-wavelength gratings

Test structures, consisting of an input coupler and an output coupler, of the fully etched grating couplers with minor sub-wavelength gratings have been fabricated through electron beam lithography at University of Washington using the JEOL JBX-6300FS E-Beam Lithography system. Test

4.2. Improved Fully Etched Grating Coupler

structures of regular fully etched grating couplers have also been fabricated for the comparison purpose. Measurement spectra of the regular fully etched grating coupler and the fully etched grating coupler with minor sub-wavelength gratings are shown in Fig. 4.10. Green curve donates the results of the regular fully etched grating coupler and the blue curve denotes the proposed fully etched grating coupler with minor sub-wavelength gratings. The extinction ratio of the ripples have been reduced from about 3dB to about 0.8 dB, by implementing the minor sub-wavelength gratings. The insertion loss of the grating coupler is also improved by about 2 dB. In addition, the bandwidth of the proposed design is largely improved, which is nearly twice as that of the regular fully etched grating coupler.

Chapter 5

Vertical Grating Coupler

All of the grating couplers mentioned so far are detuned grating couplers, which indicates the case that a small angle is employed between the incident wave to the normal of the grating surface. The small angle is introduced to avoid the second order Bragg reflection [15], but such a small angle is not desirable for a low-cost packaging process [5]. In this chapter, grating couplers for perfectly vertical coupling will be presented, and a bidirectional grating coupler for vertical coupling will be demonstrated to improve the insertion loss of the vertical grating couplers.

5.1 Regular Vertical Grating Couplers

The main drawback of the vertical grating coupler is the large second order reflection to the waveguide, which can be predicted by the Bragg condition:

$$\beta \cdot \sin\theta = \beta - m \cdot K \quad (5.1)$$

where β denotes the wave vector of the propagated mode within the grating, θ denotes the diffraction angle, m is an integer denoting the diffraction order, and $K = \lambda/\Lambda$. In the case where the first order diffraction is perfectly coupled to the fibre, i.e., $\theta = 0$ when $m = 1$, $\beta = K$, i.e., $\Lambda = \lambda/n_{\text{eff}}$, where n_{eff} denotes the effective index of the grating and λ denotes the central wavelength. In this case, the second order diffraction of the grating is in the direction of the input waveguide, i.e., $\theta = -90$ when $m = 2$. The wave vector diagram of a vertical grating coupler is shown in Fig. 5.1.

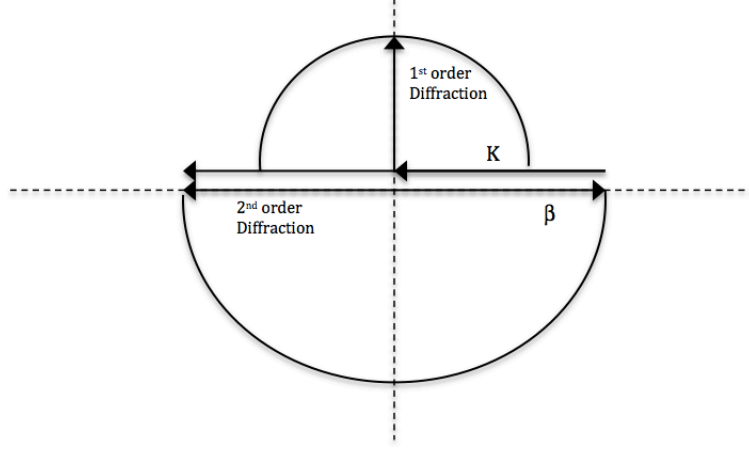


Figure 5.1: Diagram of wave vectors for vertical grating coupler

A vertical grating coupler is designed based on a SOI wafer with a $2\text{ }\mu\text{m}$ buried oxide and a 220 nm top Si layer. A shallow etch layer with an etch depth of 70 nm is used and SiO_2 is employed as the cladding for protection. The designed central wavelength is 1550 nm and FDTD simulations have been used to find the optimized design with low insertion loss and large bandwidth around the desired wavelength. The effective index of the 220 nm Si layer is 2.83 (n_{eff1}), and the effective index of the shallow etched Si layer is 2.506 (n_{eff2}). Therefore, the effective index of the grating can be calculated by $n_{\text{eff}} = n_{\text{eff1}}/2 + n_{\text{eff2}}/2$, which is 2.668 . Grating period of the desired grating can be calculated from the Bragg condition with the known effective index of the grating, which is 580 nanometers in our case. Further optimization of design parameters such as grating period and duty cycle have been done using FDTD methods. Figure 5.2 shows simulation results of the insertion loss and back reflection to waveguide of an optimized regular vertical grating coupler. The blue curve denotes the insertion loss and the green curve denotes the back reflection to the waveguide. A valley is shown in the spectrum of the insertion loss, which is caused by the large back reflection.

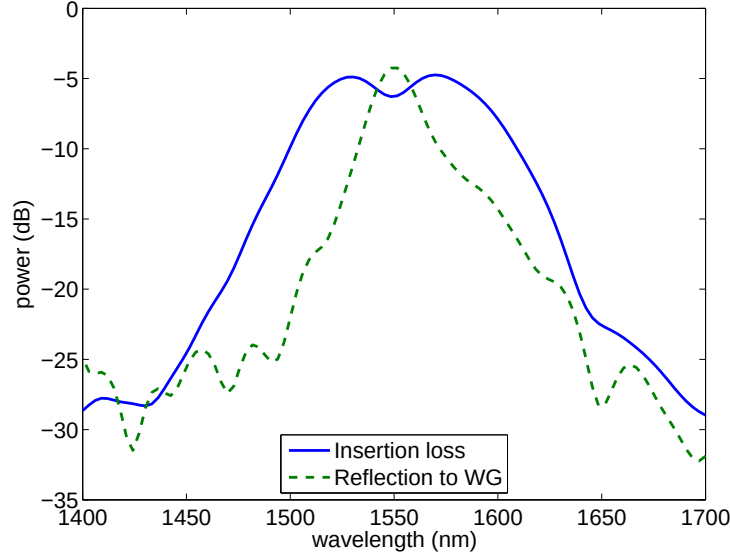


Figure 5.2: Insertion loss and back reflection of a regular vertical grating coupler

Different approaches have been employed to avoid the large second order reflection of the vertical grating couplers [4, 5, 40, 57]. Asymmetric grating structure [40] has been used to avoid the large Bragg reflection for vertical coupling. However, the extra etch depth required for fabrication increases the complexity as well as the cost. Chirped gratings were also implemented to minimize the mode mismatch between the grating and optical fibre [4, 5]. However, high fabrication accuracy is required to fabricate those chirped grating couplers. Given the fabrication accuracy of the existing technology, the stability and repeatability of the chirped gratings are still doubted. Another drawback of the chirped grating coupler is their small bandwidth. The reported 3dB bandwidth of the chirped vertical grating coupler is only about 45 nanometers. Slanted grating coupler was proposed to obtain high coupling efficiency to a vertically positioned optical fibre [57]. However, the technology required for fabrication of the slanted grating slit is too compacted to be compatible with standard CMOS technology, and therefore only suitable for prototyping.

5.2 Bidirectional Grating Coupler

5.2.1 Device Layout

The regular vertical grating couplers only couple light from one side of the grating. Theoretically, the insertion loss of such vertical grating couplers with uniform gratings can exceed 50 %. Here we introduced a bidirectional grating coupler structure for vertical grating, which breaks the theoretical limitation of regular vertical grating couplers. A schematic of the bidirectional grating coupler is shown in Fig. 5.2. Basically, it is a symmetric structure with the grating in the centre. Two-dimensional tapers are connected to both sides of the grating to convert the mode into the waveguides. Bending waveguides are employed to direct the light from both sides of the grating into a 50-50 coupler. Finally, light from both sides of the grating is combined together with a 50-50 coupler.

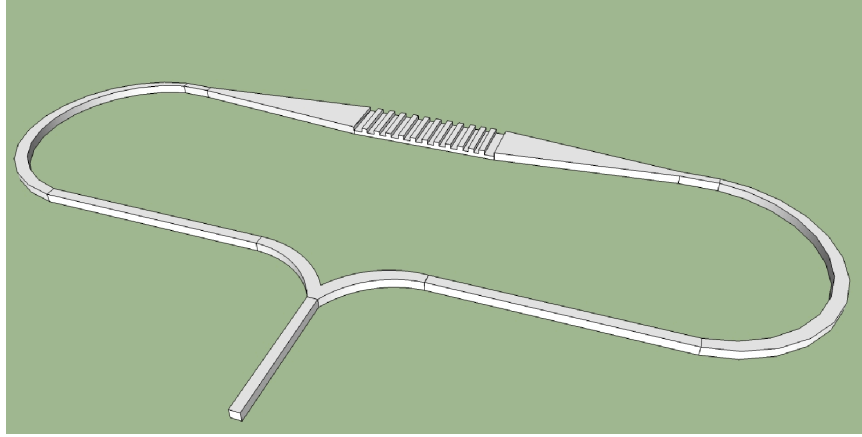


Figure 5.3: Schematic of bidirectional grating coupler for vertical coupling

Figure 5.4 shows the cross-section of the bidirectional vertical grating coupler designed in a SOI wafer with a 2 μm buried oxide and a 300 nm top Si layer. A shallow etch layer ($ed = 155\text{nm}$) has been employed by the design, and SiO_2 is used as the cladding for protection. Λ denotes the grating period and ff denotes the duty cycle of the grating coupler.

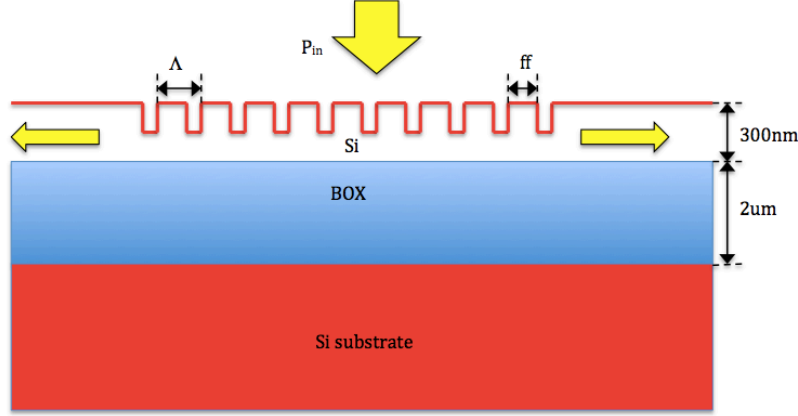


Figure 5.4: Cross section of a bidirectional grating coupler

5.2.2 Design and Simulation

FDTD method has been used to design and simulate the bidirectional grating coupler design. The grating period of the grating coupler can be predicted by the Bragg condition. Given that the effective index of the 300 nm slab waveguide is 3.052 (n_{eff1}), and the effective index of the 145 nm slab waveguide is 2.571 (n_{eff2}), the effective index of the grating (n_{eff}) can be obtained, i.e., $n_{\text{eff}} = 0.5 * n_{\text{eff1}} + 0.5 * n_{\text{eff2}}$, which is 2.812 in our case. The desired central wavelength is 1550 nanometers, so the grating period can be obtained from the Bragg condition, which is about 550 nm in our case.

2D FDTD simulations were used to further optimize the design. Three parameters, i.e., the number of grating, grating period, and the duty cycle were optimized using the Particle Swarm Optimization algorithm to achieve a design with low insertion loss and large bandwidth. The number of grating period determines the length of the coupling region. As the length of the grating region increases, the directionality of the grating increases, which means more power can be diffracted to the fibre, but extra scattering loss will be introduced as the light propagates from the grating to the waveguide. On the other hand, shorter grating region can reduce the scattering loss, but the directionality of the grating will decrease. The grating period along with

5.2. Bidirectional Grating Coupler

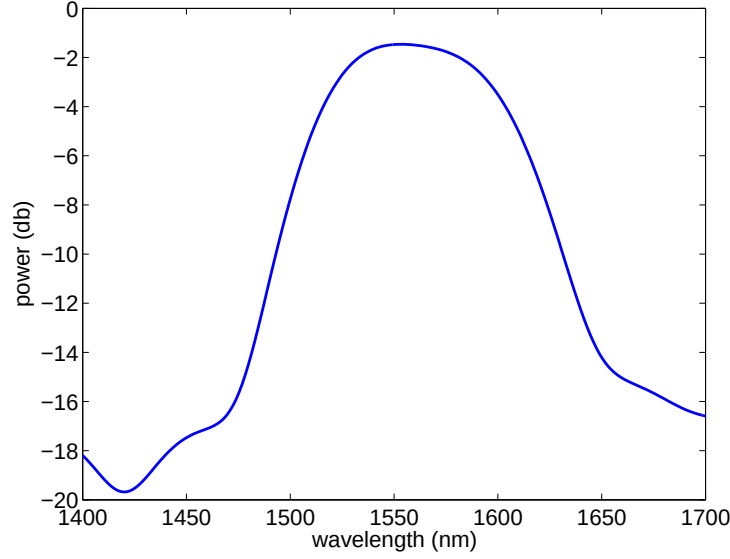


Figure 5.5: The insertion loss of the optimized bidirectional grating coupler

the duty cycle of the grating determines the central wavelength of the grating coupler. The optimized design has a grating number of 13, a grating period of 580 nanometers, and a duty cycle of 240 nanometers. The insertion loss of the optimized bidirectional vertical grating couple is shown in Fig. 5.5. The insertion loss is calculated with the assumption that the taper and Y junction are lossless. The assumption holds because theoretically a taper with a length of 200 micron can achieve lossless conversion from 12 μm grating to a 500 nm wire waveguide[50], and compact Y junction with low loss has been demonstrated [42, 61]. The insertion loss of the optimized grating coupler is only -1.5 dB, which is much smaller than of that of the regular vertical grating coupler. The 3dB bandwidth of the bidirectional grating coupler is more than 100 nanometers, which is more than twice of the value reported in [5].

The optical waves in different sides of the grating behave like two arms of a MZI. The insertion loss shown in Fig. 5.5 can be only obtained when the incident wave from the fibre impinges in the middle of the grating coupler. In the case when the incident wave is off from the centre of the grating, optical

5.2. Bidirectional Grating Coupler

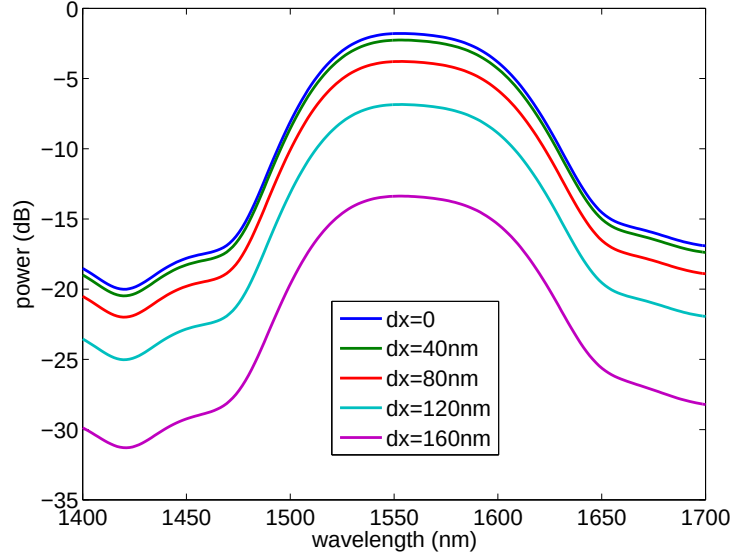


Figure 5.6: Insertion of the bidirectional vertical grating coupler as function of wavelength for different offset

waves from the two arms interfere with each other at the junction of the 50-50 coupler. Depending on the deviation of the incident wave, either constructive interference or destructive interference can be obtained at the 50-50 coupler. Figure 5.6 shows the simulation results of the bidirectional grating coupler with various deviation values. The deviation of the input wave, i.e., dx shown in Fig. 5.7, is defined as the distance from centre of the input fibre core to the centre of the grating. The effective index of the channel waveguide used in our case is about 2.4, and the central wavelength of the input wave is 1550nm. According to the phase condition between the propagated wave in the two arm, i.e., $n_{\text{eff}} \cdot 2dx = \lambda/2$, the first destructive interference will be obtained when $dx = 160\text{nm}$. Fig. 5.6 shows the simulation results for various deviation values. As the deviation, dx , goes beyond 160 nm, the insertion loss of the vertical grating coupler will be somewhere in between the maximum value and the minimum value. From the simulation results shown in Fig. 5.6, we note that the insertion loss of the bidirectional vertical grating coupler is very sensitive to the position of the input fibre, which results in

5.2. Bidirectional Grating Coupler

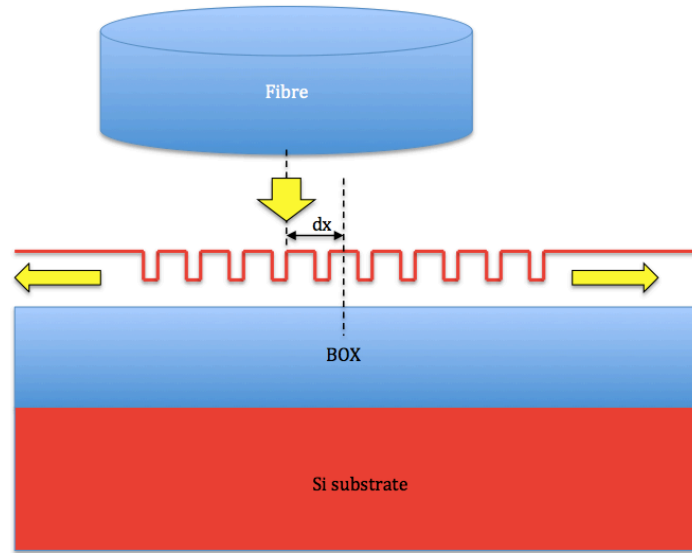


Figure 5.7: Schematic of a bidirectional grating coupler with incident wave off from the centre of the grating.

difficulties for alignment of the grating during the measurement. Highly accurate motor controllers are required for the alignment of the bidirectional grating couplers during the measurement.

Chapter 6

Discussion And Future Work

In this thesis, we mainly presented three different works:

1. A universal design methodology has been presented to design shallow etched grating couplers, which accommodates various etch depths, silicon thicknesses, and cladding materials.
2. A fully etched grating coupler with minor sub-wavelength gratings has been demonstrated to improve the insertion loss, back reflection, and the bandwidth of the regular fully etched grating coupler.
3. A bidirectional grating coupler for vertical coupling has been presented.

The point of the universal design methodology is that it simplifies the design process of a grating coupler. The traditional way of designing a grating coupler involves brute-force simulations on various design parameters, which requires a lot of computational memory and time. Using the universal design method, time-consuming simulations have been replaced by simple analytical calculations. We also implemented the design process in Pyxis to generate the mask layout of the desired grating coupler with the input parameters determined by the fabrication process and the designer. The analytical calculation is implemented into the scripts so no simulation is ever needed to design a grating coupler using our design methodology. The performance of the grating couplers designed by this method is comparable to the published results, but with a much simplified design flow for system level designers. Table 6.1 shows the comparison between our measurement results and published results. Since the performance of the grating coupler is highly dependent on the wafer type and fabrication process, we only

listed the published results which used similar wafer types and fabrication processes with us. Future work can be done to add a correction coefficient to the existing model, thereby further improving the wavelength mismatch.

	Pl.	IL	Bandwidth	Process
2006 [48]	TE	-5.1 dB	40nm (1dB)	220nm Si, 1um BOX
2012 [60]	TE	-4.4dB	45nm (1.5dB)	220nm Si, 2um BOX
Ours	TE	-4.8dB	45nm(3dB)	220nm Si, 2um BOX

Table 6.1: Comparison of the published results and the result of the grating coupler generated by the universal design methodology

A fully etched grating coupler with minor sub-wavelength grating has also been proposed to improve the performance of regular fully etched grating coupler. Due to the fully etched slots, the coupling strength of the fully etched grating couplers is very large, which makes high-efficiency coupling more difficult to achieve. In addition, the thickness of buried oxide of the most commonly used SOI wafer is 2 μm , which is not ideal for efficient coupling. Large back reflection to the waveguide leads to strong oscillation ripples in the transmission. However, the large back reflection of the fully etched grating coupler has been successfully reduced by employing the minor sub-wavelength gratings between the major gratings. A larger bandwidth has also been achieved by the proposed design. Further effort can be applied to add chirped grating period to the design, therefore the mode overlap between the grating and the optical fibre can be further improved to reduce the insertion loss.

In addition, a bidirectional grating coupler for vertical coupling was proposed to reduced the fabrication complexity and improve the insertion loss at the same time. The published vertical grating coupler designs employed non-standard fabrication processes such as slanted grating [57] and extra etch depth [40], which are only suitable for prototyping. The proposed bidirectional grating coupler only requires uniform gratings with a single shallow etch layer, which is compatible with the current CMOS technology. The simulated insertion loss of this optimized design is as low as -1.46 dB

with a 3dB bandwidth of more than 100nm. The 3 dB bandwidth of the bidirectional vertical grating coupler is much larger than the value reported by other research groups. Further work can be done to design a better Y branch with lower insertion loss.

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Appendix A

FDTD code to generate universal grating coupler model

```
## Universal Grating Coupler Based on EIM and Bragg Condition
## definition of variables:
## period: grating pitch;
## duty_cycle: the unetched part of a grating in the longitudinal direction;
## fill factor: the ratio of duty_cycle and period;
## etch depth: the length of the etched part of the grating in the vertical
direction;
## neff: effective index of the grating region;
## ne1: effective index of the unetched region;
## ne2: effective index of the etched region;
## Si_thickness: the thickenss of the unetched silicon;

##Initialization;
Si_thickness=d1=0.22e-6;
etch_depth=0.075e-6;
theta=20;
n_c=1.44;
lambda=1.58e-6;
switchtolayout;
redrawoff;
selectall;
```

```
delete;

# draw silicon substrate;
addrect;
set('name','Si_sub');
set('material','Si (Silicon) - Palik');
set('x',0);
set('x span', 40e-6);
set('y',-3e-6);
set('y span',2e-6);

#draw burried oxide;
addrect;
set('name','BOX');
set('material','SiO2 (Glass) - Palik');
set('x',0);
set('x span', 40e-6);
set('y',-1e-6);
set('y span',2e-6);
set('override mesh order from material database',true);
set('mesh order',3);
set('alpha',0.3);

#draw waveguide;
addrect;
set('name','WG');
set('material','Si (Silicon) - Palik');
set('x min',0);
set('x max', 20e-6);
set('y',0.11e-6);
set('y span',0.22e-6);
```



```
# add simulation region;
addfddtd;
set('x max',5e-6);
set('x min',-14e-6);
set('y min',-2.3e-6);
set('y max',1e-6);
if(n_c>1)
{
select('BOX');
set('y max',1.93e-6);

select('FDTD');
set('y max',3e-6);
}

## calculating neff for fundamental TE mode in fully etched WG;
addmode;
set('name','mode');
set('x',2e-6);
set('y',0.5*d1);
set('y span',1e-6);
set('direction','Backward');
set('center wavelength',lambda);
set('wavelength span',0.3e-6);
set('mode selection','fundamental mode');
select('WG');
set('y',0.5*d1);
set('y span',d1);
select("mode");
updatesourcemode;
neff1=getresult('mode','neff');
ne1=neff1.neff;
```

Appendix A. FDTD code to generate universal grating coupler model

```
# calculating neff for fundamental TE mode in shallow etched WG;
select('WG');
set('y',0.5*(d1-etch_depth));
set('y span',(d1-etch_depth));
select("mode");
set('y',0.5*(d1-etch_depth));
clearsourcedata;
updatesourcemode;
neff2=getresult('mode','neff');
ne2=neff2.neff;

# calculating neff assuming ff is 50%;
neff=0.5*(ne1+ne2);
?neff;

# change WG strcture to original thickness;
select('WG');
set('y',0.11e-6);
set('y span',0.22e-6);

select('mode');
set('y',0.11e-6);

## calculate the period of the grating based on neff;
period=lambda/(neff-sin((theta/180)*pi));
?period;
duty_cycle=0.5*period;

## draw uniform GC
addrect;
set('name','GC_base');
set('x max',0);
set('material','Si (Silicon) - Palik');
```

```
set('x min', -20e-6);
set('y min', 0);
set('y max', d1-etch_depth);
for (i=0:40)
{
    addrect;
    set('name', 'GC_tooth');
    set('x min', -period-i*period);
    set('x max', -duty_cycle-i*period);
    set('y', 0.5*d1);
    set('y span', d1);
}
selectpartial('tooth');
set('material', 'Si (Silicon) - Palik');
selectpartial('GC');
addtogroup('GC');

## add Gaussian mode
select('mode');
set('enabled', 'false');
addgaussian;
set('name', 'gaussian');
set('x', -4.5e-6);
set('x span', 16e-6);
set('direction', 'Backward');
set('polarization angle', 90);
set('angle theta', theta);
set('center wavelength', lambda);
set('wavelength span', 0.1e-6);
set('waist radius w0', 4.5e-6);
set('distance from waist', 10e-6);
if (n_c>1)
{
```

```
set('y',2.5e-6);
}
else
{
set('y',0.5e-6);
}

## add monitor;
addpower;
set('name','r');
set('monitor type','Linear Y');
set('x',3e-6);
set('y',0.5*d1);
set('y span',1e-6);

addpower;
set('name','u');
set('monitor type','Linear X');
set('x',-4.5e-6);
set('x span',20e-6);
if(n_c>1)
{
set('y',2.8e-6);
}
else
{
set('y',0.8e-6);
}

addpower;
set('name','d');
set('monitor type','Linear X');
set('x',-4.5e-6);
```

```
set('x span',20e-6);
set('y',-2.1e-6);
run;
switchtolayout;
redrawoff;
selectall;
delete;

# draw silicon substrate;
addrect;
set('name','Si_sub');
set('material','Si (Silicon) - Palik');
set('x',0);
set('x span', 40e-6);
set('y',-3e-6);
set('y span',2e-6);

#draw burried oxide;
addrect;
set('name','BOX');
set('material','SiO2 (Glass) - Palik');
set('x',0);
set('x span', 40e-6);
set('y',-1e-6);
set('y span',2e-6);
set('override mesh order from material database',true);
set('mesh order',3);
set('alpha',0.3);

#draw waveguide;
addrect;
set('name','WG');
set('material','Si (Silicon) - Palik');
```

```
set('x min',0);
set('x max', 20e-6);
set('y',0.11e-6);
set('y span',0.22e-6);

# add simulation region;
addfddtd;
set('x max',5e-6);
set('x min',-14e-6);
set('y min',-2.3e-6);
set('y max',1e-6);
if(n_c>1)
{
select('BOX');
set('y max',1.93e-6);
select('FDTD');
set('y max',3e-6);
}

## calculating neff for fundamental TE mode in fully etched WG;
addmode;
set('name','mode');
set('x',2e-6);
set('y',0.5*d1);
set('y span',1e-6);
set('direction','Backward');
set('center wavelength',lambda);
set('wavelength span',0.3e-6);
set('mode selection','fundamental TE mode');
select('WG');
set('y',0.5*d1);
set('y span',d1);
select("mode");
```

Appendix A. FDTD code to generate universal grating coupler model

```
updatesourcemode;
neff1=getresult('mode','neff');
ne1=neff1.neff;

# calculating neff for fundamental TE mode in shallow etched WG;
select('WG');
set('y',0.5*(d1-etch_depth));
set('y span',(d1-etch_depth));
select("mode");
set('y',0.5*(d1-etch_depth));
clearsourcedata;
updatesourcemode;
neff2=getresult('mode','neff');
ne2=neff2.neff;

# calculating neff assuming ff is 50%;
neff=0.5*(ne1+ne2);
?neff;

# change WG strcture to original thickness;
select('WG');
set('y',0.11e-6);
set('y span',0.22e-6);
select('mode');
set('y',0.11e-6);

## calculate the period of the grating based on neff;
period=lambda/(neff-sin((theta/180)*pi));
?period;
duty_cycle=0.4*period;
## draw uniform GC
addrect;
set('name','GC_base');
```

```
set('x max',0);
set('material','Si (Silicon) - Palik');
set('x min', -20e-6);
set('y min',0);
set('y max',d1-etch_depth);
for (i=0:40)
{
addrect;
set('name','GC_tooth');
set('x min',-period-i*period);
set('x max',-duty_cycle-i*period);
set('y',0.5*d1);
set('y span',d1);
}
selectpartial('tooth');
set('material','Si (Silicon) - Palik');
selectpartial('GC');
addtogroup('GC');

## add Gaussian mode
select('mode');
set('enabled','false');
addgaussian;
set('name','gaussian');
set('x',-4.5e-6);
set('x span', 16e-6);
set('direction','Backward');
set('polarization angle',0);
set('angle theta',theta);
set('center wavelength',lambda);
set('wavelength span',0.3e-6);
set('waist radius w0',4.5e-6);
set('distance from waist',10e-6);
```



```
if (n_c>1)
{
set('y',2.5e-6);
}
else
{
set('y',0.5e-6);
}

## add monitor;
addpower;
set('name','r');
set('monitor type','Linear Y');
set('x',3e-6);
set('y',0.5*d1);
set('y span',1e-6);
addpower;
set('name','u');
set('monitor type','Linear X');
set('x',-4.5e-6);
set('x span',20e-6);
if(n_c>1)
{
set('y',2.8e-6);
}
else
{
set('y',0.8e-6);
}
addpower;
set('name','d');
set('monitor type','Linear X');
set('x',-4.5e-6);
```

Appendix A. FDTD code to generate universal grating coupler model

```
set('x span',20e-6);  
set('y',-2.1e-6);  
run;
```

Appendix B

Pyxis code for universal grating coupler design

```
function EBeam_UGC_double()
{
    local device = $get_device_iobj();
    local wl = $get_property_value(device,"wl");
    local etch_depth = $get_property_value(device,"etch_depth");
    local Si_thickness = $get_property_value(device,"Si_thickness");
    local incident_angle = $get_property_value(device,"incident_angle");
    local wg_width = $get_property_value(device,"wg_width");
    local n_cladding = $get_property_value(device,"n_cladding");
    local pl = $get_property_value(device,"pl");
    build_EBeam_UGC_double(wl,etch_depth,Si_thickness,incident_angle,
        wg_width,n_cladding,pl);
}

function build_EBeam_UGC_double( wl:number {default=1.55},
etch_depth:number {default=0.07}, Si_thickness:number {default=0.22},
incident_angle:number {default=20},wg_width:number {default=0.5},
n_cladding:number{default=1.44},pl:number {default=1})
{

    //Save Original user settings
    local selectable_types_orig = $get_selectable_types();
    local selectable_layers_orig = $get_selectable_layers();
    local autoselect_orig = $get_autoselect();
```

```
//Set up selection settings
$set_selectable_types(@replace, [@shape, @path, @pin, @overflow,@row,
@property_text, @instance, @array, @device, @via_object, @text, @region,
@bisector, @channel, @slice], @both);
$set_selectable_layers(@replace, ["0-4096"]);
$set_autoselect(@true);

// calculate the neff of the grating region
// local for neff calculation

local point=1001;
local ii=0;
local jj=0;
local kk=0;
local mm=0;
local nn=0;
local n0=n_cladding;
local n1 = 0;
local n2 = 3.473;
local n3 = 1.444;
local delta=n0-n3;
local t = Si_thickness;
local t_slot=t-etch_depth;
local k0 = 2*3.14159/wl;
local b0 = $create_vector(point-1);
local te0 = $create_vector(point-1);
local te1 = $create_vector(point-1);
local tm0 = $create_vector(point-1);
local tm1 = $create_vector(point-1);
local h0 = $create_vector(point-1);
local q0 = $create_vector(point-1);
local p0 = $create_vector(point-1);
```

```
local qbar0 = $create_vector(point-1);
local pbar0 = $create_vector(point-1);
local mini_TE=0;
local index_TE=0;
local mini_TE1=0;
local index_TE1=0;
local mini_TM=0;
local index_TM=0;
local mini_TM1=0;
local index_TM1=0;
local nTE=0;
local nTE1=0;
local nTM=0;
local nTM1=0;
local ne=0;

//...// calculating neff for 0.22 silicon layer

    if( delta<0)
    {
        n1=n3;
    }
    else
    {
        n1=n0;
    }

    for(ii=0;ii<point-1;ii=ii+1)
    {
        b0[ii]= n1*k0+(n2-n1)*k0/(point-10)*ii;
    }

    for(jj=0;jj<point-1;jj=jj+1)
```

```
{
  h0[jj] = sqrt( abs(pow(n2*k0,2) - pow(b0[jj],2)));
  q0[jj] = sqrt( abs(pow(b0[jj],2) - pow(n0*k0,2)));
  p0[jj] = sqrt( abs(pow(b0[jj],2) - pow(n3*k0,2)));
}

for(kk=0;kk<point-1;kk=kk+1)
{
  pbar0[kk] = pow(n2/n3,2)*p0[kk];
  qbar0[kk] = pow(n2/n0,2)*q0[kk];
}

//calculating neff for TE mode

if (pl==1)
{
  for (nn=0;nn<point-1;nn=nn+1)
  {
    te0[nn] = tan( h0[nn]*t )-(p0[nn]+q0[nn])/h0[nn]/
      (1-p0[nn]*q0[nn]/pow(h0[nn],2));
    te1[nn] = tan( h0[nn]*t_slot )-(p0[nn]+q0[nn])/h0[nn]/
      (1-p0[nn]*q0[nn]/pow(h0[nn],2));
  }

  local abs_te0=abs(te0);
  local abs_te1=abs(te1);
  mini_TE=$vector_min(abs_te0);
  mini_TE1=$vector_min(abs_te1);
  index_TE=$vector_search(mini_TE,abs(te0),0);
  index_TE1=$vector_search(mini_TE1,abs(te1),0);
  nTE=b0[index_TE]/k0;
  nTE1=b0[index_TE1]/k0;
```

```
do
{
    abs_te0[index_TE]=100;
    mini_TE=$vector_min(abs_te0);
    index_TE=$vector_search(mini_TE,abs(te0),0);
    nTE=b0[index_TE]/k0;
}
while ( nTE<2 || nTE>3);

//    $message($format("%4.5f",nTE));

do
{
    abs_te1[index_TE1]=100;
    mini_TE1=$vector_min(abs_te1);
    index_TE1=$vector_search(mini_TE1,abs(te1),0);
    nTE1=b0[index_TE1]/k0;
}
while ( nTE1<2 || nTE1>3);

//    $message($format("%4.5f",nTE1));

ne=0.55*nTE+0.45*nTE1;
$message($format("%4.5f",ne));
}

//calculating neff for TM mode
else
{
    for (mm=0;mm<point-1;mm=mm+1)
    {
        tm0[mm] = tan(h0[mm]*t)-h0[mm]*(pbar0[mm]+qbar0[mm])/
        (pow(h0[mm],2)- pbar0[mm]*qbar0[mm]);
```

```
tm1[mm] = tan(h0[mm]*t_slot)-h0[mm]*(pbar0[mm]+qbar0[mm])/
(pow(h0[mm],2)-pbar0[mm]*qbar0[mm]);

}

local abs_tm0=abs(tm0);
local abs_tm1=abs(tm1);
mini_TM=$vector_min(abs(tm0));
mini_TM1=$vector_min(abs(tm1));
index_TM=$vector_search(mini_TM,abs(tm0),0);
index_TM1=$vector_search(mini_TM1,abs(tm1),0);
nTM=b0[index_TM]/k0;
nTM1=b0[index_TM1]/k0;

do
{
    abs_tm0[index_TM]=100;
    mini_TM=$vector_min(abs_tm0);
    index_TM=$vector_search(mini_TM,abs(tm0),0);
    nTM=b0[index_TM]/k0;
}
while ( nTM<1.5 || nTM>3);

// $message($format("\%4.5f",nTM));

do
{
    abs_tm1[index_TM1]=100;
    mini_TM1=$vector_min(abs_tm1);
    index_TM1=$vector_search(mini_TM1,abs(tm1),0);
    nTM1=b0[index_TM1]/k0;
}
while ( nTM1<1.5 || nTM1>3);
```



```
// $message($format("%4.5f",nTM1));

ne=0.45*nTM+0.55*nTM1;
$message($format("%4.5f",ne));
}

//...// end of calculating neff

// Start of grating drawing
local period=-1*w1/(sin(rad(incident_angle))*1-ne);
local duty_cycle=0;

$message($format("%4.5f",period));
if (pl==1)
{
duty_cycle=period*0.5;
$message($format("%4.5f",duty_cycle));
}
else
{
duty_cycle=period*0.45;
$message($format("%4.5f",duty_cycle));
}

local segnum=75; //segment number of one grating curve
local seg_points = segnum+1;
local arc_vec = $create_vector(2*seg_points+2);
local i = 0;
local j = 0;
local x_r=0;
local y_r=0;
local x_l=0;
local y_l=0;
```

```
local m1_x=0;
local m1_y=0;
local m2_x=0;
local m2_y=0;
local nf=1.44;
local e=nf*sin(rad(incident_angle))/ne;
local gc_number=$round(21/period);
local angle_e=62;
local N=$round(18*(1+e)*ne/wl)+1;
local gap=period-duty_cycle;

for(j=0;j<gc_number;j=j+1)
{
    for(i=0;i<seg_points;i=i+1)
    {
        x_r=(N*wl/(ne*(1-e*cos(rad(180-angle_e/2+angle_e/segnum*i))))+
        j*period)*cos(rad(180-angle_e/2+angle_e/segnum*i));
        y_r=(N*wl/(ne*(1-e*cos(rad(180-angle_e/2+angle_e/segnum*i))))+
        j*period)*sin(rad(180-angle_e/2+angle_e/segnum*i));
        arc_vec[i] = [x_r,y_r];

        m1_x=(N*wl/(ne*(1-e*cos(rad(180+angle_e/2))))+
        duty_cycle/2+j*period)*cos(rad(180+angle_e/2));
        m1_y=(N*wl/(ne*(1-e*cos(rad(180+angle_e/2))))+
        duty_cycle/2+j*period)*sin(rad(180+angle_e/2))-0.1;
        arc_vec[seg_points] = [m1_x,m1_y];

        x_l=(N*wl/(ne*(1-e*cos(rad(180+angle_e/2-angle_e/segnum*i))))+
        gap+j*period)*cos(rad(180+angle_e/2-angle_e/segnum*i));
        y_l=(N*wl/(ne*(1-e*cos(rad(180+angle_e/2-angle_e/segnum*i))))+
        gap+j*period)*sin(rad(180+angle_e/2-angle_e/segnum*i));
        arc_vec[seg_points+i+1] = [x_l,y_l];
```

```
m2_x=(N*wl/(ne*(1-e*cos(rad(180+angle_e/2-angle_e/segnum*i))))+
gap/2+j*period)*cos(rad(180+angle_e/2-angle_e/segnum*i));
m2_y=(N*wl/(ne*(1-e*cos(rad(180+angle_e/2-angle_e/segnum*i))))+
gap/2+j*period)*sin(rad(180+angle_e/2-angle_e/segnum*i))+0.1;
arc_vec[2*seg_points+1] = [m2_x,m2_y];
}

angle_e = angle_e-0.7;
$add_shape(arc_vec,'SiEtch1');

}

$add_shape([[ -40,13.5],[0,1/2*wg_width],[0,-1/2*wg_width],[-40,-13.5]], 'Si');
//$unselect_all(@nofilter);
$add_shape([[ -1, -wg_width/2],[0, wg_width/2]], "Si", @internal);

//Restore original user settings
$set_selectable_types(@replace, (selectable_types_orig[0]==void)?
[]:selectable_types_orig[0],selectable_types_orig[1]);
$set_selectable_layers(@replace, selectable_layers_orig);
$set_autoselect(autoselect_orig);

}

function EBeam_UGC_double_parameters( layer:optional number {default=1},
wl:optional number {default=1.55}, etch_depth:optional number {default=0.07},
Si_thickness:optional number {default=0.22},incident_angle:optional number
{default=20},wg_width:optional number {default=0.5},n_cladding:optional number
{default=1.44},pl:optional number {default=1})
{ return [ ["wl",$g(wl)],["etch_depth",$g(etch_depth)],["Si_thickness",
$g(Si_thickness)],["incident_angle",$g(incident_angle)],
["wg_width",$g(wg_width)],["n_cladding",$g(n_cladding)],["pl",$g(pl)] ]; }
```